

THE EARLY CAREER AND THE SOURCES OF LIFETIME INEQUALITY*

Paulo Lins
Clemson University

Xiaonan Ma

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ABSTRACT

How much of lifetime inequality is determined by the time workers enter the labor market? The answer depends on when measurement begins. We estimate a life-cycle model with endogenous hours, learning-by-doing, match quality, and frictional labor markets, tracking workers from labor-market entry. Initial conditions explain 26 percent of lifetime earnings inequality at entry, but 46 percent five years later, implying that decompositions that start years after entry partly reclassify inequality produced during workers' early career as predetermined heterogeneity. Most of this early-career rise in earnings inequality reflects sorting: human capital and match quality become increasingly correlated over the first years of work. Sorting arises because human capital affects labor-market outcomes beyond wages – it raises job-finding rates, lowers separation risk, and shifts workers toward better offer distributions – so skill accumulation translates endogenously into better matches. The early career is therefore not a prelude to the life cycle; it is a period in which a substantial share of lifetime inequality is produced.

Keywords: Life-Cycle Models, Labor Supply, Human Capital Accumulation

JEL - Classification: D15, J22, J24

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1 Introduction

How much of lifetime inequality is determined by the time workers enter the labor market? Standard life-cycle decompositions often take a chronological age cutoff as the base period, typically in early adulthood (Huggett, Ventura, and Yaron, 2011; Ozkan, Song, and Karahan, 2023; Bick, Blandin, and Rogerson, 2024). But by then, many workers have already accumulated experience, searched across jobs, experienced nonemployment, and sorted into better or worse matches. If these early-career outcomes are shaped by endogenous labor supply, human-capital accumulation, and search frictions, then starting the decomposition after labor-market entry can reclassify inequality produced during the school-to-work transition as predetermined heterogeneity.

This concern is substantive because the early career is not a passive period. Hours worked rise sharply, employment grows, wages increase, and workers move across jobs. These patterns are difficult to interpret through either a standard life-cycle labor-supply model or a standard frictional-search model alone. Human-capital accumulation makes current work more valuable because it raises future productivity (Heckman, 1976; Imai and Keane, 2004; Fan, Seshadri, and Taber, 2024). Search frictions make early mobility valuable because workers move out of nonemployment and into better matches (Topel and Ward, 1992; Neal, 1999). Distinguishing these forces matters for understanding both labor-supply incentives and the formation of lifetime inequality.

This paper studies these forces in a life-cycle model with an endogenous work-week, learning-by-doing, match quality, and frictional labor markets. The model is built around two state variables that summarize workers' labor-market position: human capital and match quality. Workers accumulate human capital through learning-by-doing, with hours worked entering the human-capital production function. Match quality evolves through random search in a frictional labor market, consistent with models in which early-career wage growth reflects job mobility and improved matches (Bagger, Fontaine, Postel-Vinay, and Robin, 2014; Menzio, Telyukova, and Visschers, 2016). Human capital affects not only the wage paid on the current job, as in standard life-cycle models of human-capital accumulation and labor supply (Heckman, 1976; Imai and Keane, 2004), but also job-finding rates, separation rates, and the mean of the offer distribution. We refer to these additional effects as the transition-rate and offer-quality channels. The model also includes borrowing constraints and persistent heterogeneity in the disutility of work; the latter helps discipline hours dispersion, a key feature of life-cycle labor supply

emphasized by [Bick et al. \(2024\)](#).

The measurement strategy is central. We estimate the model by indirect inference using NLSY79 and NLSY97 data on high-school graduates, with profiles indexed by quarters since labor-market entry rather than by age. We identify labor-market entry with high-school graduation and restrict attention to workers whose highest degree is a high-school diploma and who do not return to school. Period zero in the model and in the data, therefore, corresponds to the same economic moment: the start of working life. Each profile bin contains workers with the same potential experience rather than the same age, avoiding the composition problem that arises when workers enter the labor market at different ages.

The first result is that the timing of measurement substantially changes the interpretation of lifetime inequality. Initial assets, human capital, match quality, and preference type explain 26.4 percent of lifetime earnings inequality when measured at labor-market entry, but 45.6 percent when measured five years later. Within the same model, moving the base period from entry to year 5 raises the explained share by 19 percentage points. The year-5 base period corresponds to age 23 for the average worker in our sample, making it comparable to the base period in [Huggett et al. \(2011\)](#).¹ State variables observed several years after entry already embed early-career shocks, labor-supply choices, human-capital accumulation, and match sorting. Starting later, therefore, reclassifies some of the inequality produced during the school-to-work transition as initial heterogeneity.

The second result is that early-career wage inequality grows primarily through sorting between human capital and match quality. Mean wage growth reflects both skill accumulation and movement toward better matches: over the first 20 years, human capital contributes 0.326 log points and match quality contributes 0.322 log points. Wage dispersion follows a different logic. The covariance between human capital and match quality moves from negative at entry to positive after five years and accounts for most of the increase in wage variance. Match-quality dispersion in isolation falls over the early career. Thus, wage inequality rises not because skill dispersion or match dispersion widens in isolation, but because workers with higher human capital increasingly hold better matches.²

¹[Huggett et al. \(2011\)](#) use age 23 as the base period and report that initial conditions account for roughly 60 percent of lifetime earnings inequality. At the comparable year-5 base period, our estimate is 45.6 percent, while the entry-based estimate is 26.4 percent. Thus, differences relative to their estimate reflect both model structure and the timing of measurement; the 19.2 percentage-point gap between our own year-5 and entry-based estimates isolates the timing effect within our framework.

²Like the lifetime-inequality decomposition above, the wage-dispersion decomposition is sensitive to the base period. Subsection [5.2](#) reports the decomposition from entry and from five years

The third result is that human-capital accumulation is the dominant source of early-career labor-supply incentives, but only about half of this incentive operates through wages. We measure dynamic incentives using the ratio of the marginal rate of substitution to the wage, as in (Imai and Keane, 2004; Keane and Rogerson, 2015). A static model predicts a ratio of one; values above one indicate that workers supply hours beyond the static optimum because work raises future productivity or improves future labor-market prospects. A search economy without human-capital accumulation generates essentially no dynamic wedge. A human-capital economy in which human capital affects only wages generates about half of the baseline dynamic wedge. The remaining incentive comes from transition-rate and offer-quality channels: higher human capital raises job-finding rates, lowers separation risk, and shifts workers toward better offer distributions.

Together, the results show that the school-to-work transition is not a short prelude to the life cycle. It is a period in which labor supply, job mobility, human-capital accumulation, and match sorting jointly produce a substantial share of lifetime inequality. Frameworks that begin after this transition or that restrict human capital to affect wages alone risk misclassifying endogenous early-career dynamics as predetermined heterogeneity.

Related Literature This paper contributes to literatures on lifetime inequality, early-career labor-market dynamics, and life-cycle labor supply. First, it relates to quantitative work decomposing lifetime inequality into the roles of initial conditions and subsequent shocks (Huggett et al., 2011; Griffl, 2021; Ozkan et al., 2023). Existing decompositions typically initialize workers at a chronological age in early adulthood. We show that this timing matters because state variables observed several years after labor-market entry already embed early-career shocks, labor-supply choices, human-capital accumulation, and match sorting. Starting the decomposition after the school-to-work transition reclassifies part of the transition as initial heterogeneity. By aligning the model and data at labor-market entry, we separate inequality present at entry from inequality produced during the first years of work.

The closest paper on early-career inequality is Kaplan (2012), who shows that unemployment is central for understanding the decline in hours inequality early in the life cycle. We share the focus on young workers, hours dispersion, and early-career labor-market frictions. Kaplan studies hours inequality in an annual incomplete-markets model in which unemployment acts as an idiosyncratic labor-

after entry; starting later shifts the apparent driver of wage-inequality growth from sorting toward human-capital dispersion.

supply wedge. We study how the school-to-work transition shapes wage dispersion and lifetime inequality. To do so, we measure workers from labor-market entry rather than from a fixed age. Our model distinguishes employment, unemployment, and job-to-job transitions, and allows hours to affect future human capital and labor-market prospects.

Second, the paper relates to work on early-career wage growth, job mobility, and match quality. Work beginning with [Topel and Ward \(1992\)](#) and [Neal \(1999\)](#) emphasizes that young workers move across jobs and occupations as they search for better matches. Quantitative search models build on this idea to decompose wage growth into experience, tenure, and match components ([Bagger et al., 2014](#); [Menzio et al., 2016](#); [Gorry, 2016](#)). In these models, experience affects wages, transitions, or information about match quality, but labor supply does not determine the rate of human-capital accumulation. We make this margin endogenous: hours worked raise future human capital, and human capital affects wages, separation risk, job-finding rates, and the quality of future offers.

Third, the paper relates to life-cycle models of labor supply with human-capital accumulation. In this tradition, current hours are an investment because they raise future wages, creating incentives to work more early in life ([Heckman, 1976](#); [Weiss, 1986](#); [Imai and Keane, 2004](#); [Wallenius, 2011](#); [Fan et al., 2024](#)). We extend this logic to a frictional labor market. Human capital affects not only the wage conditional on a job, but also the process through which workers find, keep, and improve jobs. The return to current work includes a labor-market-opportunity component in addition to the standard wage channel. Quantitatively, this distinction is important: a wage-only human-capital model generates only about half of the dynamic labor-supply incentive in the baseline economy.

Finally, related work on the school-to-work transition studies how schooling, occupational choice, training, and credit constraints shape labor-market entry ([Keane and Wolpin, 1997](#); [Lochner and Monge-Naranjo, 2011](#); [Flinn and Mullins, 2015](#)). We take labor-market entry as the starting point and study how inequality evolves afterward, as hours choices, job mobility, human-capital accumulation, and wage growth are jointly determined. Our focus on young workers also connects to work showing that early-career workers play a central role in aggregate labor-market dynamics ([Shimer, 2001](#); [Jaimovich, Pruitt, and Siu, 2013](#)) and to [Gorry \(2013\)](#), who studies youth unemployment in a search model with experience accumulation. The paper also relates to work emphasizing extensive-margin labor supply, fixed costs, and heterogeneity over the life cycle. [Erosa, Fuster, and Kambourov \(2016\)](#) show that fixed costs and preference heterogeneity are important for accounting for life-

cycle labor supply, while [Bick et al. \(2024\)](#) show that differences in lifetime hours are an important source of lifetime earnings inequality. Our model incorporates heterogeneity in labor disutility and learning ability, as in [Bick et al. \(2024\)](#), but embeds these forces in a framework with job-switching costs and explicit employment, unemployment, and job-to-job transitions.

Roadmap

Section 2 describes the life-cycle model with human capital accumulation and job search frictions. Section 3 describes the data and our measurement choices, including labor market entry and the construction of quarters-since-graduation profiles. Section 4 describes our estimation strategy. We also use a statistical decomposition of the growth into its components. Section 5 reports the quantitative results. Section 6 concludes.

2 Model

The model is built to answer one question: why do hours rise early in the life cycle? Our mechanism is that hours are an investment. Working more today raises future human capital, and the return to that investment depends on both current match quality and expected mobility. A worker in a good match earns more today, but also has less reason to move. A worker in a poor match expects future job changes, which lowers the value of accumulating match-specific human capital. Hours, therefore, respond not only to the current wage but also to the effect of current work on future wages, job stability, and access to better outside offers.

To capture this mechanism, the model combines learning-by-doing with a frictional labor market. Human capital grows through work experience. Workers search both in and out of employment, and jobs differ in match quality. Human capital and match quality jointly shape wages, offer arrival rates, and separation risk, so both state variables affect the return to current hours through several margins at once. The model also allows for saving, borrowing constraints, and heterogeneity in the disutility of work.

2.1 Setup

Time is discrete and measured in months to capture the dynamics of job transitions and separations.

Preferences Workers live for a finite horizon $t = 0, 1, \dots, T$. In each period, they choose consumption c_t and, when employed, hours worked $h_t \in [0, 1]$. Lifetime utility is

$$E_0 \sum_{t=0}^T \beta^t \left(\ln(c_t) - \xi_t \frac{h_t^{1+\frac{1}{\gamma}}}{1+\frac{1}{\gamma}} \right), \quad \beta \in (0, 1).$$

The parameter $\xi_t > 0$ governs the disutility of work, while $\gamma > 0$ controls its curvature. We allow ξ_t to vary across workers and over the life cycle. This heterogeneity helps the model match dispersion in hours, but it is not the main force behind rising average hours.

Labor Market At each date, a worker is either unemployed or employed in a match of quality m_t . Unemployed workers receive benefits $\bar{b}$ and search for jobs. Employed workers choose hours and may either remain in their current match or move to a new one when an outside offer arrives. Search is frictional in both states.

All transition rates use logistic functions of human capital and, where relevant, match quality, so that they remain in $[0, 1]$ for all states. This guarantees that continuation terms are well-defined expected values while allowing state variables to shift rates smoothly.

Each period, an unemployed worker receives a match offer with arrival rate $\lambda^U(s_t)$ that depends on his stock of human capital s_t ,

$$\lambda^U(s_t) = \frac{1}{1 + \exp\left(\lambda_0^U - \rho_{\lambda s} \ln(s_t)\right)}.$$

The parameter λ_0^U determines the baseline job-arrival rate, while $\rho_{\lambda s}$ determines how job-finding varies with human capital. Thus, more skilled workers receive job offers more frequently.

Match quality offers are drawn from an exogenous distribution. The cumulative distribution function (CDF) of match offers, $F_{q^U}(s)(m)$, is log-normal:

$$F_{q^U}(s)(m) = \Phi\left(\frac{\log(m) - \mu^U(s)}{\sigma^U(s)}\right),$$

where $\mu^U(s) = \mu_0^U + \mu_1^U s$ and $\ln \sigma^U(s) = \sigma_0^U + \sigma_1^U s$ are the mean and log standard deviation of the underlying normal distribution, and $\Phi(\cdot)$ denotes the standard normal CDF. If a worker rejects the offer, he remains unemployed; if he accepts it, he begins the next period employed at the accepted match. Human capital shifts both the mean and the spread of the offer distribution, so more skilled workers draw better matches on average. While unemployed, human capital depreciates at rate δ_u per period.

An employed worker is characterized by current human capital s_t and match quality m_t . Labor income equals the wage per unit of labor times hours worked, where the wage rate is

$$w(s_t, m_t) = w_0 s_t^{\rho_{ws}} m_t^{\rho_{wm}}.$$

The parameter w_0 determines the wage level, while ρ_{wm} and ρ_{ws} govern the sensitivity of wages to match quality and human capital.

Each period, an employed worker receives an outside offer with arrival rate $\lambda^E(s_t, m_t)$,

$$\lambda^E(s_t, m_t) = \frac{1}{1 + \exp\left(\lambda_0^E - \rho_{\lambda s} \ln(s_t) - \rho_{\lambda m} \ln(m_t)\right)}.$$

Here, λ_0^E determines the baseline arrival rate, while $\rho_{\lambda s}$ and $\rho_{\lambda m}$ determine how it varies with human capital and match quality. Match offers are drawn from an exogenous distribution distinct from that faced by unemployed workers. The CDF of match quality, $F_{q^E}(s)(m)$, is log-normal:

$$F_{q^E}(s)(m) = \Phi\left(\frac{\log(m) - \mu^E(s)}{\sigma^E(s)}\right),$$

where $\mu^E(s) = \mu_0^E + \mu_1^E s$ and $\ln \sigma^E(s) = \sigma_0^E + \sigma_1^E s$ are the mean and log standard deviation of the underlying normal distribution, and $\Phi(\cdot)$ denotes the standard normal CDF.

In addition, an employed worker faces exogenous separation risk $\pi(s_t)$, which declines with human capital:

$$\pi(s_t) = \frac{1}{1 + \exp\left(\pi_0 - \rho_{\pi s} \ln(s_t)\right)}.$$

The parameter π_0 determines the baseline separation rate, while $\rho_{\pi s} < 0$ governs how separation varies with human capital. Thus, more human capital reduces the risk of job loss. When separated, workers begin the next period unemployed. While

employed, human capital evolves according to the law of motion described below.

Human Capital Employed workers accumulate human capital through a learning-by-doing process that depends on current human capital s_t , match quality m_t , and hours worked h_t ,

$$\mathcal{H}(s_t, m_t, h_t, \alpha_t) = \alpha_t s_t^{\rho_{ss}} m_t^{\rho_{sm}} h_t^{\rho_{sh}} .$$

The parameter $\alpha_t > 0$ determines the efficiency of learning by doing, while ρ_{ss} , ρ_{sm} , and ρ_{sh} are the elasticities of human capital accumulation with respect to current human capital, match quality, and hours worked. We allow workers to differ in learning ability, as described in more detail below.

To capture the idea that part of accumulated human capital is specific to the current job or occupation, we assume that workers lose human capital when they change jobs, either voluntarily or through separation.³ We model this loss as a one-time adjustment cost that is convex in current human capital,

$$\phi(s_t) = B s_t^\nu .$$

Here, $B > 0$ determines the level of the loss and $\nu > 1$ its curvature. The convexity implies that the loss rises more than proportionally with accumulated human capital.

The law of motion for human capital is therefore

$$s_{t+1} = \begin{cases} (1 - \delta_e) s_t + \mathcal{H}(s_t, m_t, h_t, \alpha_t), & \text{if no job change ,} \\ (1 - \delta_e) s_t + \mathcal{H}(s_t, m_t, h_t, \alpha_t) - \phi(s_t), & \text{if job change or separation ,} \end{cases}$$

where δ_e is the depreciation rate while employed. Thus, hours raise future human capital through learning by doing, but job changes reduce the amount of accumulated capital carried into the next period. For unemployed workers, human capital depreciates at rate δ_u , with no new accumulation.

Budget Constraint Both employed and unemployed workers can save and borrow using a one-period risk-free asset with net return r . Borrowing is subject to the constraint $a_{t+1} \geq \bar{a}_t$, where $\bar{a}_t$ reflects a natural debt limit based on the present

³This assumption is consistent with evidence that specialized knowledge is imperfectly transferable across jobs and occupations. [Hernandez Martinez, Holter, and Pinheiro \(2025\)](#) show that workers whose pre-displacement human capital is more specialized experience larger earnings losses after displacement.

value of future income while unemployed and receiving unemployment benefits. For quantitative reasons, we allow the borrowing limit to be tighter than this natural debt limit. The per-period budget constraint is

$$c_t + a_{t+1} = \begin{cases} w(s_t, m_t)h_t + (1+r)a_t & \text{if employed,} \\ \bar{b} + (1+r)a_t & \text{if unemployed.} \end{cases}$$

Retirement Workers retire at age $T_R < T$. During retirement, they no longer supply labor and do not receive labor income. Instead, they receive common retirement income Ret . The budget constraint then becomes

$$c_t + a_{t+1} = \text{Ret} + (1+r)a_t, \quad \text{for } t \geq T_R.$$

Retirees cannot borrow against future retirement income, so the borrowing constraint tightens to $a_t \geq 0$ for $t \geq T_R$. The terminal condition is $a_{T+1} = 0$.

Initial Conditions and Type Heterogeneity Workers begin life with initial assets, human capital, match quality, and employment status. We assume that assets, human capital, and match quality follow a jointly log-normal distribution, while initial employment status is drawn from a Bernoulli distribution with unemployment probability p_{u_0} .

Workers also differ in their disutility of labor, ξ_t , and in the efficiency of human capital accumulation, α_t . The disutility parameter evolves over the life cycle according to an AR(1) process in logs,

$$\log \xi_{t+1} = (1 - \rho_\xi)\mu_\xi + \rho_\xi \log \xi_t + \varepsilon_{j,t}, \quad \varepsilon_{j,t} \sim N(0, \sigma_\varepsilon^2),$$

where μ_ξ is the unconditional mean, $\rho_\xi \in (0, 1)$ is the persistence parameter, and $\sigma_\varepsilon = \sigma_\xi \sqrt{1 - \rho_\xi^2}$ is the innovation standard deviation implied by the unconditional variance σ_ξ^2 . The initial distribution of ξ_t is log-normal with mean μ_ξ and standard deviation $\sigma_{\xi,0} < \sigma_\xi$, so dispersion increases over the life cycle as workers receive idiosyncratic shocks.

The learning-efficiency parameter α_t is allowed to vary systematically with labor disutility. Specifically,

$$\log \alpha_t = C_0^\alpha + C_1^\alpha \log \xi_t,$$

where C_0^α and C_1^α govern the relationship between learning efficiency and labor disutility. We assume this relationship is negative, so workers who dislike work more are also less efficient at accumulating human capital. This generates a reinforcing mechanism: workers with high disutility supply less labor, accumulate less human capital through experience, and learn less efficiently on the job.

2.2 Workers' Problem and Value Functions

An employed worker at time t chooses consumption c_t , savings a_{t+1} , hours worked h_t , and, when an outside offer arrives, whether to remain in the current job or move to a new employer. The value function depends on age t , assets a_t , human capital s_t , current match quality m_t , and current labor-disutility type ξ_t :

$$\begin{aligned} V_t^W(a_t, s_t, m_t, \xi_t) = & \max_{c_t, a_{t+1}, h_t} \left\{ u(c_t) - v(h_t) \right. \\ & + \beta \pi(s_t) \mathbb{E}_{\xi'} U_{t+1}(a_{t+1}, s'_{t+1}, \xi') \\ & + \beta \left(1 - \pi(s_t) \right) \lambda^E(s_t, m_t) \mathbb{E}_{m', \xi'} \left[\max \left\{ V_{t+1}^W(a_{t+1}, s_{t+1}, m_t, \xi'), V_{t+1}^W(a_{t+1}, s'_{t+1}, m', \xi') \right\} \right] \\ & \left. + \beta \left(1 - \pi(s_t) \right) \left(1 - \lambda^E(s_t, m_t) \right) \mathbb{E}_{\xi'} V_{t+1}^W(a_{t+1}, s_{t+1}, m_t, \xi') \right\}, \end{aligned}$$

subject to the budget constraint, the borrowing constraint, the hours restriction $0 < h_t < 1$, and the law of motion for human capital. Here, s_{t+1} and s'_{t+1} denote next-period human capital without and with a job transition, respectively. We omit the explicit functional forms of these laws of motion because they follow directly from the previous subsection.

Expectations are taken over next-period labor disutility, which also determines next-period learning efficiency. When an outside offer arrives, expectations also integrate over the match-offer distribution. An employed worker accepts a new offer when the value of the new match exceeds the value of remaining in the current one, taking into account the human-capital loss associated with switching. The optimal hours choice reflects not only the within-period tradeoff between consumption and leisure, but also the effect of current hours on future earnings, offer arrival rates, separation risk, and the stock of human capital carried into the future.

An unemployed worker at time t chooses consumption c_t , savings a_{t+1} , and whether to accept an offer or remain unemployed. The value function depends on age t , assets a_t , human capital s_t , and current labor-disutility type ξ_t :

$$\begin{aligned} V_t^U(a_t, s_t, \xi_t) = & \max_{c_t, a_{t+1}} \left\{ u(c_t) - v(0) \right. \\ & + \beta \lambda^U(s_t) \mathbb{E}_{m', \xi'} \left[\max \left\{ V_{t+1}^U(a_{t+1}, s_{t+1}, \xi'), V_{t+1}^W(a_{t+1}, s_{t+1}, m', \xi') \right\} \right] \\ & \left. + \beta (1 - \lambda^U(s_t)) \mathbb{E}_{\xi'} V_{t+1}^U(a_{t+1}, s_{t+1}, \xi') \right\}, \end{aligned}$$

subject to the budget constraint, the borrowing constraint, and the law of motion for human capital. As usual, accepting or rejecting an offer depends on whether the current draw exceeds the reservation match quality.

A retired worker at time $t \geq T_R$ chooses consumption c_t and savings a_{t+1} . The value function depends on age t and assets a_t :

$$V_t^R(a_t) = \max_{c_t, a_{t+1}} \left\{ u(c_t) - v(0) + \beta V_{t+1}^R(a_{t+1}) \right\},$$

subject to the retirement budget constraint and the no-borrowing constraint. The terminal condition is $V_{T+1}^R(a_{T+1}) = 0$.

Within each period, workers make their choices before shocks are realized. Shocks arrive in the following order: first, employed workers face a separation shock; second, both employed and unemployed workers receive match offers; third, labor-disutility shocks are realized.

The model contains two sources of uncertainty: shocks that arrive over the life cycle and heterogeneity in initial conditions. Separation and offer arrival events are Bernoulli, with probabilities given by the transition functions defined above. Match-quality draws are log-normal, and innovations to labor disutility are also log-normal. At labor-market entry, employment status is drawn from a Bernoulli distribution with unemployment probability p_{u0} . Conditional on employment status, initial assets, human capital, and match quality are drawn from a joint log-normal distribution. Initial labor disutility is drawn from a log-normal distribution with mean μ_ξ and standard deviation $\sigma_{\xi,0}$, while initial learning efficiency is pinned down by the mapping described above.

2.3 Discussion

Human capital and match quality are the key state variables in the model. Human capital raises wages, improves labor market prospects, and lowers separation risk. Match quality affects current pay and the value of remaining in the current job relative to moving elsewhere. Together, these two variables shape the joint evolution of wages, job mobility, and labor supply over the life cycle. Preference heterogeneity plays a different role. The disutility parameter ξ_t shifts labor supply directly and, through its correlation with learning efficiency, contributes to the dispersion of human capital over time.

The human capital channel resolves a tension that arises in standard intertemporal labor supply models. Intertemporal substitution predicts that workers should work more when wages are high and less when wages are expected to rise in the future. For young workers facing upward-sloping wage profiles, this logic implies low hours early in the career and high hours later, the opposite of the pattern observed in the data (Card, 1994). Human capital accumulation reverses this implication. Because hours today raise future earnings, working more when young is an investment, as in Heckman (1976) and Imai and Keane (2004). In this sense, expected wage growth is not taken as given; it is partly created by current labor supply.

Job search and switching costs introduce a second force through the interaction between mobility and accumulation. Workers with little experience and poor matches expect more future job transitions. Because changing jobs destroys part of the human capital accumulated in the current match, the return to current hours is lower when a worker expects to leave soon.⁴ This force dampens labor supply early in the career, when match quality is low and transition rates are high. As workers accumulate experience, they have more to lose from switching, which generates endogenous lock-in and reduces mobility with age. Together, these mechanisms produce rising hours and declining job-to-job transitions over the life cycle.

The model abstracts from several features present in richer labor market frameworks. Separations are exogenous, wages are given by a reduced-form wage function rather than bargaining, and the model omits firm heterogeneity and employer learning. Retirement is also modeled in a reduced-form way. These simplifications keep the framework tractable while preserving the mechanisms central to the pa-

⁴This logic is related to Castex, Chow, and Dechter (2025), who show that anticipated automation risk reduces human-capital investment and slows wage growth. The source of risk differs: their paper studies automation risk, whereas our model studies mobility and separation risk.

per: human capital accumulation, job search, and the loss of human capital at job transitions.

3 Measurement: Labor Market Entry and Profiles

In this section, we describe the sample construction, the identification of labor market entry, and the construction of quarter-since-graduation profiles. We use these profiles to estimate the model.

3.1 Data and Sample Restrictions

We use data from the National Longitudinal Survey of Youth 1979 and 1997 (NLSY79 and NLSY97, respectively) to construct profiles of many labor-market outcomes, particularly hours worked, wages, and transition rates. The NLSY79 is a nationally representative sample of 12,686 individuals born between 1957 and 1964 who were aged 14–22 in 1979. The NLSY97 is a nationally representative sample of 8,984 individuals born between 1980 and 1984 who were aged 12–17 in 1997. In addition to their panel structure, the main advantage of these datasets is the extensive information they provide on respondents’ labor-market and educational histories.

Our empirical analysis focuses on individuals with a high school diploma as their highest educational attainment, since their labor market entry is clearly defined.⁵ We use the NLSY employment histories to construct a quarterly panel. Variables such as weekly employment status and employer identifiers are reported for each week. Other variables, including wages, hours, and job characteristics, are reported only once per survey wave for each job. These are assigned to the survey year and, by convention, mapped to all calendar weeks within that survey year. We then aggregate the weekly data to a quarterly frequency, which is appropriate for analyzing the extensive margin and for capturing the rapidly changing labor-market outcomes of young workers.

After defining the target population and analysis frequency, we impose additional sample restrictions. Table 1 lists all restrictions and their impact on sample size. We sequentially apply the following criteria: (i) completion of the 12th grade; (ii) reporting a valid high school graduation date; (iii) graduation between ages 16

⁵For individuals who did not complete high school, entry is ambiguous—occurring either after dropping out or later. Similarly, for those with education beyond high school, it is unclear whether entry occurred after high school graduation, college graduation, or the completion of professional or postgraduate degrees.

and 24 to restrict the sample to a typical life-cycle trajectory; (iv) no military service; (v) at least 10 years of panel participation in the NLSY79 or 5 years in the NLSY97; and (vi) observations available for either the first or second year following graduation. After applying all restrictions, the final sample consists of 3,341 respondents from the original 12,686 individuals in the NLSY79 and 1,563 respondents from the original 8,984 individuals in the NLSY97.⁶

	NLSY79		NLSY97	
	Ind.	% Sample	Ind.	% Sample
Initial Sample	12,686	100.0%	8,984	100.0%
Final Sample	3,341	26.3%	1,563	17.4%
<i>Restrictions applied</i>				
Completed 12th grade	6,009	47.4%	2,386	26.6%
Reported a valid graduation date	5,521	43.5%	2,112	23.5%
Graduated between ages 16 and 24	5,072	40.0%	2,024	22.5%
Did not serve in the military	3,906	30.8%	1,911	21.3%
Met panel duration requirements	3,563	28.1%	1,803	20.1%
Observed 1st or 2nd year post-graduation	3,341	26.3%	1,563	17.4%

Table 1: Sample Selection Criteria and Final Sample Size

When computing life-cycle profiles to calibrate the model, we restrict the sample to individuals with a minimum level of labor-market attachment, defined as 520-5,200 annual hours. This restriction ensures that our profiles describe the outcomes of workers with sustained attachment, rather than being driven by individuals with little or no engagement in the labor market. A similar restriction is also used by [Kaplan \(2012\)](#) and [Bick et al. \(2024\)](#) when constructing their profiles. Importantly, most of our profiles are constructed conditional on employment, while the extensive margin is captured separately in the employment rates. We also use a statistical decomposition to quantify the contribution of entry and exit to the hours profile.

3.2 Identifying Labor Market Entry

We identify labor market entry using the high school graduation date. The NLSY79 and NLSY97 are unique for this purpose: in addition to their long panel structure

⁶Appendix [A.1](#) reports descriptive statistics to evaluate potential selection concerns arising from our sample restrictions. The table compares three groups: the full NLSY79 cohort, all high school graduates, and the final analysis sample. As expected, high school graduates differ substantially from the full cohort, but the final sample closely resembles the broader population of high school graduates.

and detailed labor market questions, they collect information on each diploma received, including the date it was awarded. Graduation dates are available in the NLSY97 but need to be created in the NLSY79. To do so, we rely on two sets of questions. First, respondents report whether they received a diploma, which diploma, and the date awarded. Second, they report the highest degree attained and the date it was obtained. When conflicting dates are reported for the same diploma, we manually cross-check responses using additional information, such as the highest degree attained up to that survey wave and the reason for leaving school.

Having identified labor-market entry, we construct quarters-since-graduation profiles. These profiles are preferable to the standard birth-age profiles for two reasons. First, within a given year bin, workers in our profiles have the same potential years of experience. By contrast, in birth-age profiles, workers are heterogeneous: at the same age, some may have just entered the labor market while others already have several years of experience. Second, quarters-since-graduation profiles align more closely with life-cycle models, which typically abstract from schooling and assume that workers are “born” when they enter the labor force.

Figure 1, Panels (a) and (b), shows the distribution of labor market entry in the NLSY79 and NLSY97, respectively. Each panel presents two histograms: the distribution for individuals with a high school diploma as their highest educational attainment and, in shaded bars, the distribution for the entire sample, including those with education above or below high school. These distributions are presented before applying any sample restrictions.

Two patterns stand out. First, the distribution of high school graduation is dispersed. In the NLSY79, the mean, median, and standard deviation of labor market entry are 19.64, 18.34, and 4.40 years, respectively; in the NLSY97, they are 19.14, 18.52, and 2.39 years. Second, by ages 21 and 25, individuals in the NLSY79 have, on average, 2.13 and 5.51 years of labor market experience, respectively, while those in the NLSY97 have 1.97 and 5.34 years, respectively. These patterns underscore why (i) using age as a proxy for labor market experience and (ii) restricting analysis to workers over 25 are problematic when studying young workers. Our approach avoids these issues by identifying entry based on high school graduation dates and constructing quarters-since-graduation profiles.

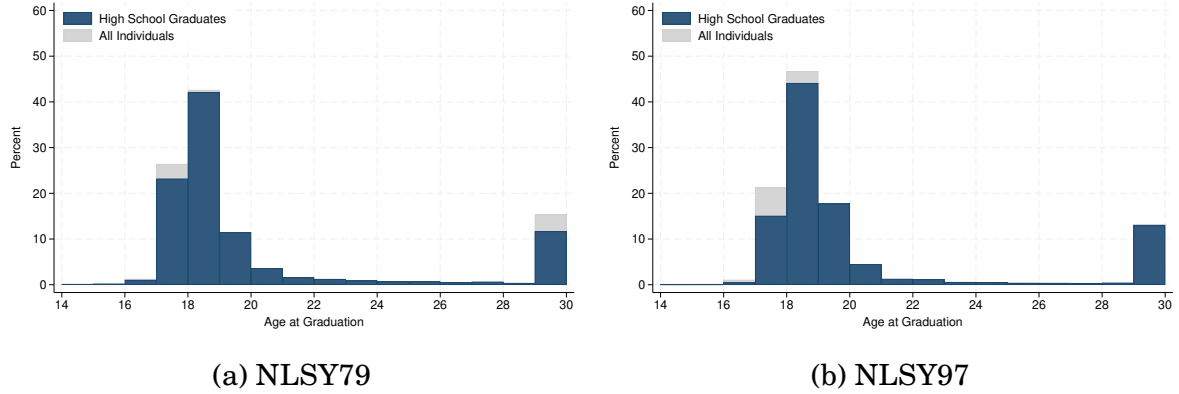


Figure 1: High School Graduation Distribution

Note: The figure plots the age distribution at high school graduation for individuals in the NLSY79 and NLSY97 samples. The lighter bars represent all individuals, while the darker bars highlight those whose highest educational attainment is a high school diploma. Ages below 14 and above 30 are grouped. For the NLSY79, it is not possible to distinguish between GED certificates and high school diplomas; both are treated equivalently. For the NLSY97, when both GED and high school graduation dates are available, the earlier date is used.

3.3 Measurement

We use eleven quarterly profiles to estimate the model. They comprise four mean outcomes – wages, employment rates, hours worked, and cumulative number of jobs – their corresponding cross-sectional standard deviations, and three labor-market transition rates.

Mean Profiles We construct four mean profiles by aggregating weekly observations to quarterly frequency. Let i index individuals and q index quarters since high school graduation.

- *Employment rate*: The fraction of weeks in quarter q that individual i spends employed, averaged across individuals. By construction, this measure lies between zero and one. In the data, non-employment includes both unemployment and non-participation.
- *Hours worked*: Total hours worked in quarter q , conditional on positive employment. We focus on hours in the main job, though respondents may hold multiple jobs within a week. We explore the difference between hours worked in the main job and a broader measure that accounts for hours worked across all jobs in an appendix. Hours are divided by 168 (the maximum weekly hours multiplied by 13) to bound them between 0 and 1.

- *Wages*: The average real hourly wage across all employed weeks in quarter q , conditional on employment. We construct real wages using the CPI. Wages are normalized to their first-quarter values, so the wage profile begins at 1.
- *Cumulative jobs*: The count of distinct jobs held by individual i up to the end of quarter q . Unlike the other profiles, this is a stock variable measured at quarter's end rather than a flow averaged over the quarter. The cumulative jobs profile is shifted so that its initial value equals one.

Standard Deviation Profiles For each mean profile, we compute the corresponding cross-sectional standard deviation. The standard deviations of hours and wages are computed conditional on positive employment and are adjusted for differences in units using logarithms. The standard deviation of employment captures dispersion in the fraction of weeks worked across individuals within a quarter.

Transition Rates We construct three transition rates from week-to-week changes in labor market status: job-to-job, job-to-unemployment, and unemployment-to-job. For each transition type, we count transitions occurring between consecutive weeks and divide by the number of individuals at risk in the origin state. We define a job-to-job transition as a direct change in job identifier between two consecutive employed weeks. A separation is a transition from employment to non-employment (unemployment or out of the labor force). Job finding is the transition from non-employment to employment.

Weekly hazard rates are converted to quarterly rates by compounding over thirteen weeks:

$$\lambda^{\text{quarter}} = 1 - (1 - \lambda^{\text{week}})^{13}.$$

We then apply a four-quarter moving average to smooth the series. Directly measured transition rates may be inconsistent with observed employment dynamics due to time-aggregation bias or measurement error. To ensure internal consistency, we adjust the separation rate following [Shimer \(2012\)](#). In particular, we use measured employment stocks to back out implied transition rates that satisfy an accounting identity. The employment identity $e_t = (1 - s_t)e_{t-1} + f_t(1 - e_{t-1})$ implies a separation rate

$$s_t = \frac{e_{t-1} + f_t(1 - e_{t-1}) - e_t}{e_{t-1}},$$

where e_t is the employment rate and f_t is the job-finding rate. We use this implied separation rate in estimation rather than the directly measured rate.

4 Estimation

Our estimation strategy proceeds in two steps. First, we assign values to some parameters that are well established in the literature or directly observable in the data. Second, we estimate the remaining parameters by matching empirical and model-based profiles.

4.1 Mapping Model Variables to the Data

Before describing the estimation procedure, we clarify how model objects correspond to their empirical counterparts. The wage function $w(s_t, m_t)$, which depends on human capital and match quality, maps to the hourly wage rate in the NLSY. Hours worked in the model, $h_t \in [0, 1]$, correspond to the average workweek in the data, which we normalize by 168 hours to ensure the same bounds. The employment rate in the model is computed as the fraction of months worked in a quarter, consistent with the data definition based on weeks worked. In the model, non-employment encompasses both unemployment and non-participation; given our sample restriction to workers with 520 to 5,200 annual hours, this approximation is reasonable. Finally, cumulative jobs in the model are measured by counting transitions – both separations to unemployment and job-to-job moves – and correspond to the number of distinct jobs held in the data, both recorded as end-of-period stocks.

4.2 Externally Parameters

We calibrate 15 parameters externally, as summarized in Table 2.

Demographics and Institutions. We set the total number of periods to $T = 840$, corresponding to a 70-year life cycle at monthly frequency. Retirement occurs at period $T_R = 540$, implying 45 years of active working life followed by 25 years of retirement. This corresponds to a worker who enters the labor market at age 20 and retires at age 65. The monthly interest rate is set to $r = 0.0025$, equivalent to approximately 3 percent per year. The wage scale w_0 is normalized to unity.

Retirement income is determined by a replacement rate $\kappa = 0.18$. This value reflects the product of three components: a 60 percent replacement ratio for high school graduates, consistent with U.S. Social Security; the approximate fraction of total weekly hours devoted to work (roughly 0.3); and our wage normalization.

Parameter	Description	Value
<i>Demographics and Institutional</i>		
T	Total life periods (70 years, monthly)	840
T_R	Retirement period (45 years of work life)	540
r	Interest rate (monthly, 3% annual)	0.0025
w_0	Wage scale normalization	1.00
κ	Replacement rate for retirement income	0.18
$\bar{b}$	Unemployment benefit	0.12
<i>Preferences</i>		
β	Discount factor (monthly)	0.994
γ	Frisch elasticity of labor supply	0.55
<i>Initial Asset Conditions</i>		
p_{A_0}	Probability of zero initial assets	0.16
μ_{A_0}	Mean of log initial assets	1.26
σ_{A_0}	Std. dev. of log initial assets	0.64
$\bar{a}$	Borrowing limit	4.54
<i>Labor Market</i>		
p_{e_0}	Initial employment rate	0.75
δ_e	Human capital depreciation (employed)	0.0025
$\rho_{\lambda m}$	Offer rate elasticity w.r.t. match quality	0.00

Table 2: Externally Calibrated Parameters

Unemployment benefits follow the same logic: $\bar{b} = 0.12 = 0.4 \times 0.3 \times 1.0$, where the first term is the unemployment insurance replacement rate, approximately 40 percent of prior earnings for this population.

Preferences The utility of consumption follows a logarithmic specification. The curvature parameter for the disutility of labor is set to $\gamma = 0.55$, consistent with micro estimates of the Frisch elasticity around 0.5 for high school-educated workers.⁷ The discount factor is set to $\beta = 0.994$ per month, implying an annual discount factor of approximately 0.93. This relatively low value implies impatient workers who tend to save less and discount the future heavily.

Initial Asset Conditions Initial asset holdings are calibrated from the 1983 wave of the Survey of Consumer Finances (SCF), restricting the sample to high school-educated household heads aged 25 or younger. Net worth is expressed in 1993 dollars and scaled by average monthly entry earnings from the NLSY79, com-

⁷A low value of γ implies a highly convex disutility of labor, which helps stabilize hours worked over the life cycle after an initial period of growth. By contrast, other studies estimating life-cycle models with endogenous labor supply and human capital accumulation obtain higher values: Imai and Keane (2004) report $\gamma \approx 3.82$, while Wallenius (2011) find $\gamma \approx 1.13$.

puted as mean hourly wages multiplied by $0.3 \times 168 \times 4.3$, where 0.3×168 reflects the average weekly hours in the NLSY sample and 4.3 is the number of weeks per month. The borrowing limit $\bar{a}$ is set to the first percentile of the scaled net worth distribution, yielding $\bar{a} = 4.54$. The initial cross-sectional distribution of assets is specified as a two-component mixture: a point mass at zero, reflecting the empirical share of households with no assets ($p_{A_0} = 0.16$), and a lognormal distribution for strictly positive holdings with parameters $(\mu_{A_0}, \sigma_{A_0}) = (1.26, 0.64)$ estimated by matching the mean and variance of log assets among positive observations.

Labor Market The initial employment rate p_{e_0} is set to match the average employment rate observed in the NLSY79 during the first quarter after graduation. A significant share of respondents report working while still enrolled in high school, so calibrating this rate is necessary to capture the gradual transition into full labor market attachment. Human capital depreciates at a rate $\delta_e = 0.0025$ per month while employed, roughly 3% annually.

We set the elasticity of offer arrival with respect to match quality to $\rho_{\lambda m} = 0$, so that the on-the-job arrival rate depends only on human capital. The parameters governing the baseline separation rate π_0 and the offer arrival rates λ^U and λ^E are calibrated internally within the estimation routine to match the separation rate, job-finding rate, and job-to-job transition rate in the first quarter after graduation, as measured in the NLSY. This calibration ensures that the model replicates the initial labor market dynamics before the estimated elasticity and age-decay parameters shape the subsequent life-cycle profiles. We report these parameters in the next table.

4.3 Internally Calibrated Parameters

Table 3 reports the 32 parameters estimated using our indirect inference procedure. We estimate the model parameters by minimizing the distance between simulated and empirical moments using a multi-stage global optimization procedure inspired by [Arnoud, Guvenen, and Kleineberg \(2019\)](#). In the first stage, we evaluate the objective function at 2^{12} Sobol quasi-random points spanning the parameter space, which provides systematic coverage of the high-dimensional domain. We then select the 2^4 best-performing points and use each as a starting value for local optimization via a derivative-free algorithm. In a final refinement stage, we form convex combinations of the current best estimate with each remaining local minimum and re-run local optimization from these combined starting points. This

strategy guards against convergence to local minima by repeatedly probing the region between competing candidates, and we take the overall minimum across all stages as our point estimate $\hat{\theta}$. When computing the distance between data and model moments for estimation purposes, we use an identity matrix as the weight matrix.

We compute standard errors using a bootstrap-based sandwich estimator. On the data side, we draw 100 bootstrap samples by resampling individuals with replacement within each cohort and recompute all empirical moments for each draw. The covariance matrix of these bootstrap moment vectors yields $\hat{\Omega}$, our estimate of the sampling variance of the data moments. On the model side, let $g(\theta) = \hat{m} - f(\theta)$ denote the vector of moment deviations, where $\hat{m}$ collects the empirical moments and $f(\theta)$ their model-implied counterparts. We approximate the Jacobian $\hat{J} = \partial f(\theta)/\partial \theta'$ by central finite differences, perturbing each parameter by $\pm 10^{-6}$ around $\hat{\theta}$ and re-solving the model at each perturbation. Because we minimize the unweighted sum of squared moment deviations (i.e. $W = I$), the asymptotic variance-covariance matrix of the estimator is given by the classical minimum distance sandwich formula,

$$\widehat{\text{Var}}(\hat{\theta}) = (\hat{J}'\hat{J})^{-1} \hat{J}'\hat{\Omega}\hat{J}(\hat{J}'\hat{J})^{-1},$$

and we report the square roots of its diagonal as standard errors.⁸

⁸The table also reports the transition rate levels λ^U , λ^E , and π_0 , which are calibrated within the estimation routine to match the first-quarter transition rates. When computing their standard errors, we use the full sample.

Parameter	Description	Value	Std. Error
<i>Initial Distributions</i>			
μ_{s_0}	Mean of log initial human capital	0.072	0.012
σ_{s_0}	Log std. dev. of initial human capital	-0.567	0.011
μ_{m_0}	Mean of log initial match quality	0.415	0.005
σ_{m_0}	Log std. dev. of initial match quality	-0.469	0.021
$\rho_{sm,0}$	Correlation of initial $\log s_0$ and $\log m_0$	-0.558	0.041
<i>Labor Market Parameters</i>			
λ^U	Offer arrival rate (unemployed)	0.615	0.000
λ^E	Offer arrival rate (employed)	1.909	0.000
π_0	Separation rate intercept	2.637	0.000
$\rho_{\lambda s}$	Offer rate elasticity w.r.t. human capital	0.050	0.000
$\rho_{\pi s}$	Separation rate elasticity w.r.t. human capital	2.258	0.000
$\rho_{\pi m}$	Separation rate elasticity w.r.t. match quality	0.908	0.000
ρ_{ws}	Wage elasticity w.r.t. human capital	0.301	0.000
ρ_{wm}	Wage elasticity w.r.t. match quality	0.750	0.000
<i>Human Capital Parameters</i>			
ρ_{ss}	HC accumulation elasticity w.r.t. human capital	0.003	0.000
ρ_{sm}	HC accumulation elasticity w.r.t. match quality	0.009	0.000
ρ_{sh}	HC accumulation elasticity w.r.t. hours	0.353	0.000
δ_u	HC depreciation rate (unemployed)	0.006	0.000
B	Scale parameter in $\phi(s)$	0.010	0.000
ν	Curvature parameter in $\phi(s)$	4.859	0.000
<i>Match Quality Distribution</i>			
μ_0^E	Mean intercept	-0.035	0.000
μ_1^E	Mean slope w.r.t. human capital	1.102	0.000
σ_0^E	Log std. dev. intercept	-1.548	0.000
σ_1^E	Log std. dev. slope w.r.t. human capital	0.021	0.000
μ_0^U	Mean intercept	0.153	0.000
μ_1^U	Mean slope w.r.t. human capital	-0.097	0.000
σ_0^U	Log std. dev. intercept	-2.397	0.000
σ_1^U	Log std. dev. slope w.r.t. human capital	-0.409	0.000
<i>Type Heterogeneity</i>			
μ_ξ	Mean labor disutility (levels)	78.02	0.000
σ_ξ	Log std. dev. of stationary $\log \xi$	-0.351	0.000
ρ_ξ	Persistence of $\log \xi$	0.996	0.000
$\sigma_{\xi,0}$	Log std. dev. of initial $\log \xi$	-1.029	<i>n.i.</i>
C_0^α	Mean α in levels	1.869	0.000
C_1^α	Log std. dev. of α	-1.327	0.000
<i>Age Effects</i>			
ζ_ξ	Age decay in labor disutility	-0.0003	0.000
ζ_λ	Age decay in offer arrival rate	0.0010	0.000
ζ_π	Age decay in separation rate	-0.0010	0.000
ζ_w	Age decay in wages	-0.0002	0.000
ζ_α	Age decay in HC accumulation	0.0000	0.000

Table 3: Internally Calibrated Parameters

Initial Distributions The parameters μ_{s_0} , σ_{s_0} , μ_{m_0} , σ_{m_0} , and $\rho_{sm,0}$ characterize the joint distribution of human capital and match quality at labor market entry. The negative correlation ($\rho_{sm,0} = -0.56$) implies that workers with high human capital tend to enter lower-quality matches. These parameters are mostly identified by the shapes of the first observations from the wage and hours profiles. Importantly, the negative correlation is identified by the high log-workweek dispersion observed early in workers' careers.

Labor Market Parameters The intercepts λ^U , λ^E , and π_0 are calibrated within the estimation routine to match the first-quarter transition rates observed in the data. They need to be calibrated together with other parameters, since they depend on the initial distributions of match quality and the human capital stock, as well as the elasticity parameters.

The elasticities $\rho_{\lambda s}$, $\rho_{\pi s}$, and $\rho_{\pi m}$ govern how transition rates respond to human capital and match quality. The separation rate is highly sensitive to human capital ($\rho_{\pi s} = 2.26$) and match quality ($\rho_{\pi m} = 0.91$), implying that more experienced workers and better matches face substantially lower separation risk. The offer arrival rate is almost independent of human capital ($\rho_{\lambda s} = 0.05$). These parameters are mostly identified by the shape of the transition profiles.

The elasticities ρ_{ws} and ρ_{wm} govern how wages respond to human capital and match quality. Wages load on both human capital ($\rho_{ws} = 0.30$) and match quality ($\rho_{wm} = 0.75$), with match quality playing a larger role. These parameters are mostly identified by the shapes of the profiles of mean wages and cross-sectional dispersion of wages.

Human Capital Accumulation The elasticities ρ_{ss} , ρ_{sm} , and ρ_{sh} shape the human capital production function. The elasticity with respect to the stock ($\rho_{ss} = 0.00$) generates almost no increasing returns of human capital accumulation as workers gain experience. The positive elasticity with respect to hours ($\rho_{sh} = 0.35$) implies that working more accelerates skill acquisition, providing an investment motive for early-career labor supply. Match quality plays a minor role ($\rho_{sm} = 0.01$). The depreciation rate while unemployed ($\delta_u = 0.006$) exceeds that while employed, penalizing non-employment spells. These parameters are mostly identified by the wage profile, but they also influence the hours profiles.

The scale B and curvature ν in the loss function $\phi(s) = Bs^\nu$ determine how much human capital is forfeited upon job transitions. The high curvature ($\nu = 4.86$)

implies that losses are convex in the stock: workers with more accumulated experience have more to lose. This generates endogenous lock-in, reducing job mobility as tenure grows and helping explain why job-to-job transitions decline with age. These parameters are mostly identified by the job-to-job transition rates and the number of jobs held.

Match Quality Distributions The parameters governing match quality differ between employed and unemployed workers. For employed workers, the mean of the offer distribution rises steeply with human capital ($\mu_1^E = 1.10$), so that skilled workers draw from a more favorable distribution. For unemployed workers, the dependence on human capital is now negative ($\mu_1^U = -0.10$), and the distribution is more compressed ($\sigma_0^U = -2.40$ versus $\sigma_0^E = -1.55$, both in logs). These parameters are mostly identified by the dispersion profiles.

Type Heterogeneity The disutility of labor ξ_t is highly persistent ($\rho_\xi = 0.996$) and fans out over the life cycle: initial dispersion ($\sigma_{\xi,0} = -1.03$ in logs) is smaller than stationary dispersion ($\sigma_\xi = -0.35$ in logs). The parameters that map the accumulation of human capital imply that it is negatively correlated with the disutility of labor. These parameters are identified by the dispersion in hours.

Age Effects To improve the model's fit, we allow for age-decay parameters that capture residual life-cycle variation in transition rates not explained by human capital and match quality. The decay operates multiplicatively. For example, the transition rate at age t is scaled by $(1 + \zeta)^t$ relative to its baseline value. All parameters are small, implying small residual life-cycle variation not captured by the human capital and match quality.

4.4 Targeted Moments: Model Fit

Figure 2 compares the model-implied and data profiles for average outcomes. Panels (2a) and (2b) present the employment rate and average workweek. The data show a rise in employment rates and an increase in the average workweek during the initial years of employment, before reaching a plateau. The model matches both patterns relatively well, capturing the rise and the plateau after some years in both measures. The rise in workweek is steeper in the data than in the model. Panel (2c) presents the average wage profile. The data shows a continuous upward trend that the model successfully reproduces. The model overestimates wages later in

the life cycle by 50 percentage points. Finally, panel (2d) presents the number of jobs held over time. The data shows high job mobility in the early years of the labor market, with frequent job changes occurring within the first 15 years. After this period, job mobility decreases as workers transition to longer-term positions. The model replicates this pattern, capturing the early-career job changes followed by a decline in job switching, but it also overestimates the number of jobs later in the life cycle.

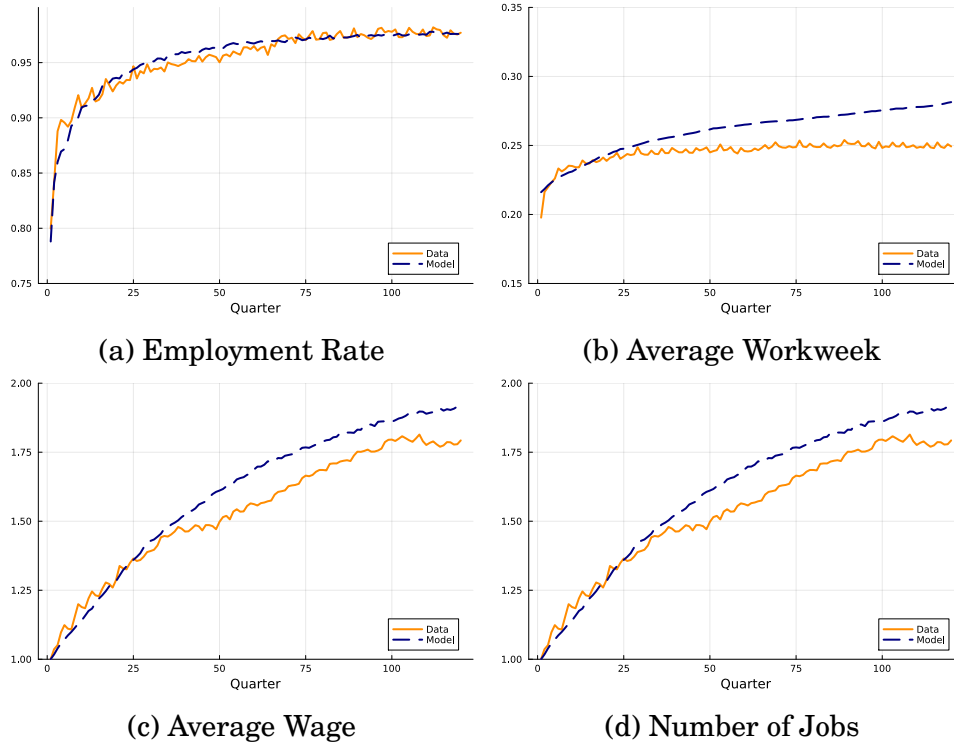


Figure 2: Comparison of Model-Implied and Data Profiles

Figure 3 compares the model-implied and data profiles for dispersion (standard deviations) of outcomes. Panels (3a) and (3b) report the cross-sectional standard deviation of the employment rate and log hours, respectively. In the data, dispersion for employment rates is larger in the early labor-market years and then stabilizes as workers sort into more persistent employment. However, the work-week displays a U-shaped pattern, as noted by Kaplan (2012). It is high early on, it decreases markedly in the first years, and starts to rise again. Importantly, the heterogeneity in preferences and its dynamics are crucial for generating this moment. Without it, hours dispersion would decrease markedly over the life-cycle.

Panel (3c) shows the standard deviation of wages, which increases over the life cycle in the data. The model also produces rising wage dispersion, consistent with mechanisms of human capital accumulation and job changes. Finally, panel

(3d) reports dispersion in the number of jobs held. The data exhibit substantial heterogeneity in early-career mobility: some workers switch frequently while others settle quickly. The dispersion in the number of jobs held continues to increase over the life cycle.

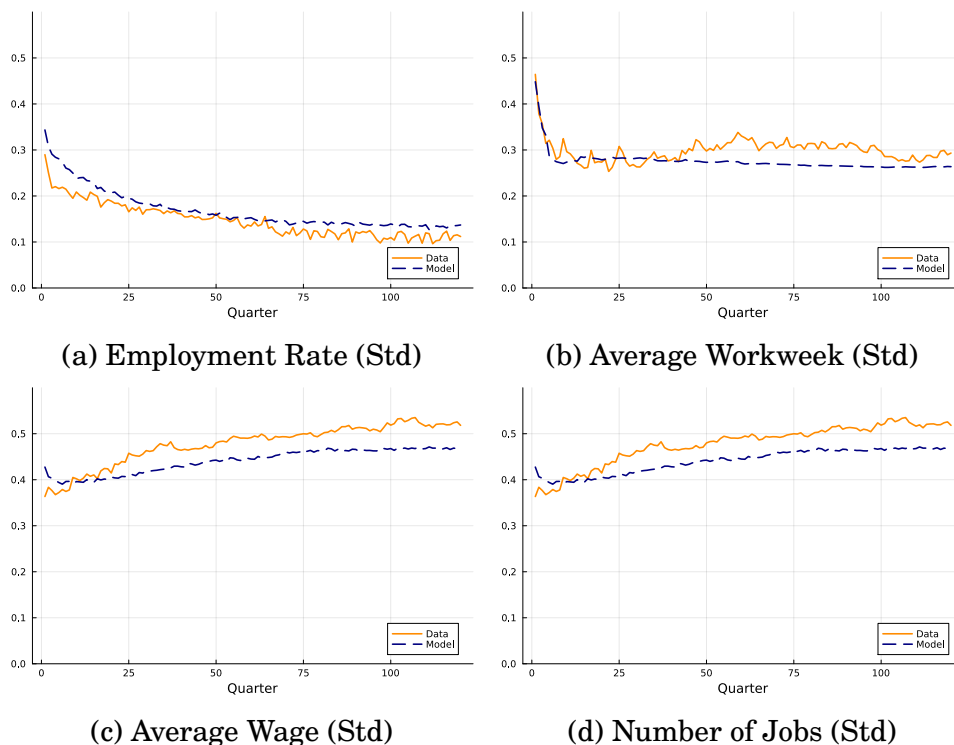


Figure 3: Comparison of Model-Implied and Data Profiles (Standard Deviations)

Finally, figure 4 reports the life-cycle profiles of key labor-market transition rates. The job-to-job (EE) transition rate is highest early on and then falls steadily, consistent with rapid early-career reallocation followed by greater job stability; the model captures the downward slope reasonably well, although it understates the very high level of EE transitions at labor-market entry. The job-to-unemployment (EU) separation rate declines with experience in both the model and the data, and overall, the model tracks the observed decline in separations. The data exhibit some spikes due to adjustments that make them consistent with other series. Finally, the unemployment-to-job (UE) rate declines over time in the data and in the model, indicating that job finding becomes less likely later in the working life.

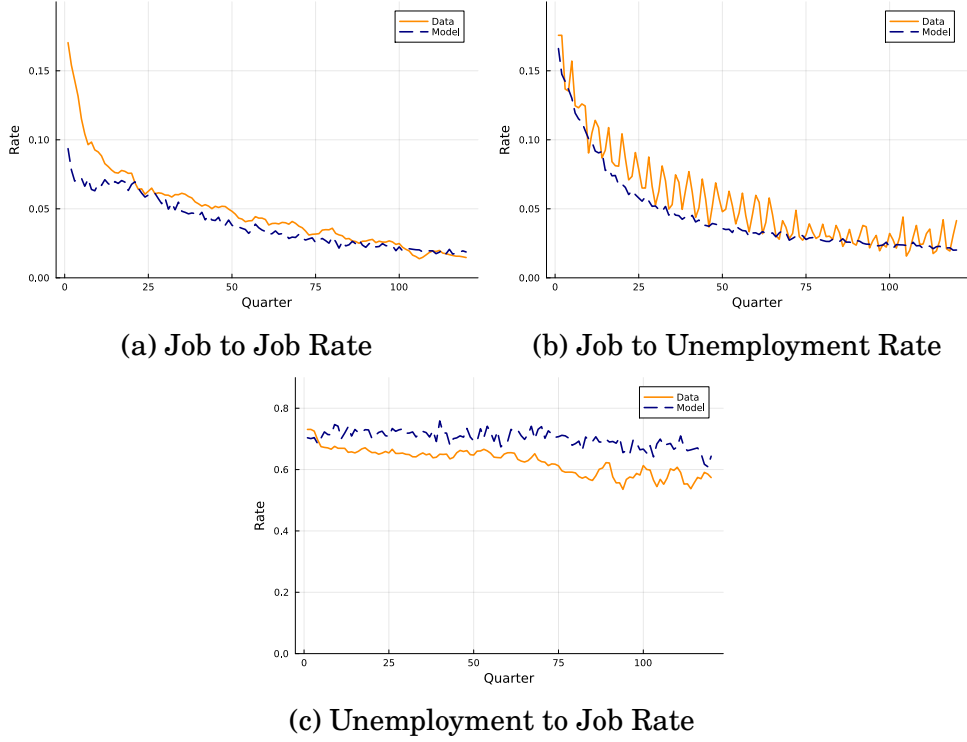


Figure 4: Comparison of Model-Implied and Data Profiles (Standard Deviations)

4.5 Untargeted Moments: Decomposing Hours into Components

Hours worked over the life cycle can rise either because individuals work more weeks per year (the extensive margin) or because they work longer weeks (the intensive margin). Many models of labor supply emphasize the extensive margin, focusing on participation as the main source of hours dynamics. In contrast, we show that the intensive margin dominates: most of the increase in hours comes from longer workweeks rather than additional weeks of employment. We apply our proposed decomposition to our quantitative life-cycle model and to the NLSY data. Importantly, this decomposition is treated as an untarget moment.

Intensive vs. Extensive Margins Let H_t denote the mean hours worked t quarters after labor-market entry. With S_t the set of individuals observed in t and $N_t = |S_t|$ the number of individuals in S_t , then

$$H_t = \frac{1}{N_t} \sum_{i \in S_t} H_{it} . \quad (1)$$

Equivalently, equation (1) can be written as a weighted average,

$$H_t = \sum_{i \in S_t} \omega_{it} H_{it}, \quad \omega_{it} = \frac{1}{N_t}.$$

Individual hours are the product of the average workweek and the number of weeks worked:

$$H_{it} = h_{it} n_{it},$$

where h_{it} is average hours per week and n_{it} is weeks worked. We compute $h_{it} = H_{it}/n_{it}$ if $n_{it} > 0$ and $h_{it} = 0$ otherwise. Following the literature, we include only workers with positive hours worked.

The life-cycle hours profile $\{H_t\}_{t=0}^T$ is the sequence of mean hours worked from entry through period T . To quantify the contributions of the intensive and extensive margins, we decompose changes in adjacent mean hours using a method similar to that of [Baily, Hulten, and Campbell \(1992\)](#) and [Blundell, Bozio, and Laroque \(2011\)](#). Specifically, the change between $t - 1$ and t can be written as

$$\begin{aligned} H_t - H_{t-1} = & \underbrace{\sum_{i \in S_{t \cap t-1}} \omega_{it} (h_{it} - h_{it-1}) n_{it-1}}_{\text{intensive margin}} \\ & + \underbrace{\sum_{i \in S_{t \cap t-1}} \omega_{it} h_{it-1} (n_{it} - n_{it-1})}_{\text{extensive margin (1): within-continuers}} \\ & + \underbrace{\sum_{i \in S_{t/t \cap t-1}} \omega_{it} h_{it} n_{it} - \sum_{i \in S_{t-1/t \cap t-1}} \omega_{it-1} h_{it-1} n_{it-1} + \sum_{i \in S_{t \cap t-1}} (\omega_{it} - \omega_{it-1}) h_{it-1} n_{it-1}}_{\text{extensive margin (2): composition effect}} \\ & + \underbrace{\sum_{i \in S_{t \cap t-1}} \omega_{it} (h_{it} - h_{it-1}) (n_{it} - n_{it-1})}_{\text{quadratic term}}. \end{aligned} \tag{2}$$

The first, second, and fourth terms use individuals observed in both periods, which we call *continuers* $S_{t \cap t-1} = \{i \mid i \in S_t \cap S_{t-1}\}$. The third term reflects sample turnover and uses *entrants* and *leavers*: individuals who appear in t but not in $t - 1$, and individuals who appear in $t - 1$ but not in t . These groups are represented by $S_{t-1/t \cap t-1}$ and $S_{t/t \cap t-1}$, respectively. Appendix B provides the step-by-step derivation.⁹

⁹Our decomposition can be interpreted as a discrete-time approximation to the total derivative, measuring how total hours would change if either the intensive or extensive margin were held constant. This approach is analogous to a Laspeyres index, in which changes are evaluated relative to a base period. Because it is a discrete approximation, second-order interaction terms do not cancel and must be explicitly included.

The first term captures the intensive margin: the change in total hours due to variation in the workweek, holding weeks worked fixed at the $t - 1$ level. The second term captures the within-continuers component of the extensive margin: the change in total hours due to variation in weeks worked, holding the workweek fixed at the $t - 1$ level. The third term captures the composition effect: the change in total hours due to entry and exit from the sample and the resulting reweighting of continuers. The fourth term captures the interaction effect: the change in total hours due to simultaneous changes in weeks worked and the workweek.

This decomposition is important because it directly reveals the relative importance of the extensive and intensive margins and thereby informs modeling choices. If the extensive margin were the primary driver of life-cycle hours growth, the intensive margin contribution would be negligible, and models of labor supply could focus exclusively on participation and weeks worked. [Fan et al. \(2024\)](#) is a recent paper that takes this route. By contrast, if the intensive margin is central, then explicitly modeling variation in the workweek becomes essential.

Finally, we track how hours accumulate over the life cycle by adding up the changes between each period:

$$H_T - H_0 = \sum_{t=1}^T (H_t - H_{t-1}) . \quad (3)$$

This gives the total increase in hours worked from entry to period T . We can also express this as a growth rate:

$$\frac{H_T - H_0}{H_0} = \frac{\sum_{t=1}^T (H_t - H_{t-1})}{H_0} . \quad (4)$$

Each component of the decomposition (2) – intensive, extensive, composition, and interaction – can be summed in the same way. Adding them up over all periods shows how much each channel contributes to the overall rise in hours worked across the life cycle.

Implementation of the Decomposition Figure 5 summarizes the main finding. It plots the contributions of the intensive, extensive (weeks), extensive (composition), and quadratic margins in green, orange, blue, and purple, respectively. The intensive margin accounts for nearly 100 additional hours over the ten years after graduation, while the extensive margin accounts for about 60 hours. The quadratic term’s contribution is negligible. The model reproduces the same pat-

tern, even though it misses some of the hours growth due to missing hours and to early-on employment rate growth in the labor market. Interestingly, the model captures the negligible impact of the quadratic term and the composition effect. The results show that life-cycle hours growth is driven primarily by longer workweeks rather than by greater participation in the data or the model. We investigate the mechanism behind the growth in the workweek in the next section.¹⁰

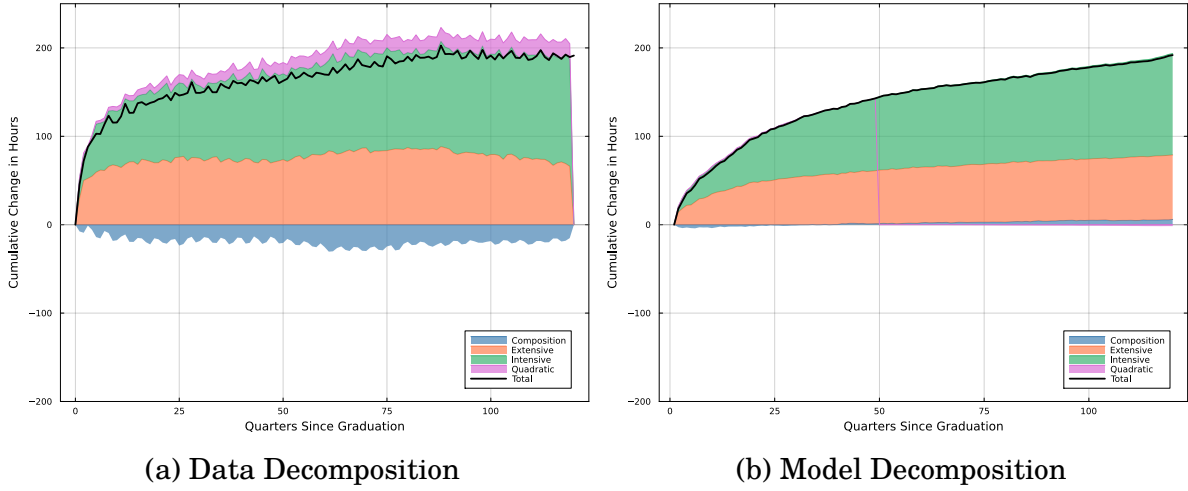


Figure 5: Intensive vs. Extensive Margin Decomposition

Table 4 helps see the magnitude. It shows the percentage contribution of each margin to cumulative growth in hours worked. We present results across different horizons: 5, 10, 15, 20, and 25 years after graduation. Row 1 reports that, five years after graduation, the intensive margin accounts for 57% of the increase in hours worked. The intensive margin is measured by the increase in the average workweek. By contrast, the extensive margin accounts for 38%, split into a 49% contribution from additional weeks worked among continuers and a -12% contribution from changes in the composition of workers in the sample. The quadratic term plays only a minor role, contributing less than -5% . Observe that, even though it is not a target, the model replicates the relative importance of each margin well. The contribution of the intensive margin increases with years of labor market experience. Fifteen years after graduation, the intensive margin accounts for 65% of the cumulative increase in hours worked, while the extensive margin accounts for 35%. Again, this dominance of the intensive margin implies that longer workweeks, not greater participation, drive the rise in hours worked after graduation.

¹⁰All results use the same sample as in Appendix A.2, pooling the NLSY79 and NLSY97 cohorts.

Table 4: Contributions of Intensive and Extensive Margins: Data vs. Model

Data				
Years	Intensive	Extensive (1)	Extensive (2)	Quadratic
5	57.0%	49.4%	-11.6%	5.2%
10	57.3%	45.0%	-9.4%	7.1%
15	59.6%	47.1%	-14.9%	8.2%
20	56.5%	45.9%	-10.2%	7.8%
Model				
5	51.8%	49.5%	-2.1%	0.8%
10	56.5%	43.1%	0.2%	0.3%
15	57.7%	41.0%	1.5%	-0.2%
20	58.2%	40.6%	1.7%	-0.5%

Note: Extensive (1) refers to within-continuers margin. Extensive (2) refers to composition margin. All values represent percentage contributions to cumulative growth up to year X.

5 Quantitative Experiments

In this section, we use the estimated model to study three related questions. First, how is lifetime inequality decomposed into initial conditions and subsequent shocks, and how does this decomposition depend on when measurement begins? Second, what drives the growth and dispersion of wages and earnings over the early career? Third, what are the underlying mechanisms – the channels through which human capital and match quality operate – that generate these patterns? The three exercises produce a unified message: the early career is a period of disproportionate consequence for both labor supply and inequality, and measurement exercises that start after the school-to-work transition miss the dynamics generated during it..

5.1 Lifetime Inequality: Initial Conditions vs Shocks

We now use the estimated model to study the sources of lifetime inequality. The exercise follows [Huggett et al. \(2011\)](#), who decompose lifetime dispersion into the share accounted for by initial conditions and the share generated by subsequent shocks. Our exercise differs in one important respect: our initial period corresponds to labor-market entry, rather than a fixed age cutoff – [Huggett et al. \(2011\)](#) use age 23. This distinction is central. If the first years after entry are a period of rapid skill accumulation, job shopping, and sorting – as the subsequent subsections document

– then an age-based decomposition treats outcomes generated during the school-to-work transition as predetermined initial conditions.

Let Y_i denote a lifetime outcome for worker i , and let X_{i0} denote the vector of initial states. We use the law of total variance,

$$\text{Var}(Y_i) = \text{Var}(\mathbb{E}[Y_i | X_{i0}]) + \mathbb{E}[\text{Var}(Y_i | X_{i0})],$$

and report the fraction

$$\frac{\text{Var}(\mathbb{E}[Y_i | X_{i0}])}{\text{Var}(Y_i)}.$$

This statistic measures the share of lifetime inequality accounted for by differences already present at the base period; the residual is attributed to subsequent shocks.

We compute the decomposition for four lifetime outcomes: lifetime earnings (the discounted sum of earnings over the working life), lifetime utility (the discounted sum of period utility over the full life cycle), lifetime wealth (lifetime earnings plus assets at the base period), and lifetime hours (total hours over the working life). We report three nested definitions of initial conditions. The first includes initial assets and human capital, (a_0, s_0) . The second adds initial employment and match quality, m_0 . The third adds the initial preference type, ξ_0 .¹¹

Table 5 reports the decomposition for two base periods: labor-market entry and five years after entry. Initial conditions at labor-market entry account for only a modest share of lifetime inequality. Under the broadest definition – assets, human capital, match quality, and preference type – initial conditions account for 26.4 percent of lifetime earnings inequality, 33.8 percent of lifetime wealth inequality, 49.2 percent of lifetime utility inequality, and 17.7 percent of lifetime hours inequality. The role of initial conditions is largest for utility because preferences enter period utility directly.

The nested columns show that match quality and preference heterogeneity each contribute substantially. In particular, the preference type and the corresponding hours results speak to Bick et al. (2024): preference heterogeneity is the most important addition to the initial state for explaining lifetime hours inequality, raising the explained share from 10.0 to 17.7 percent. Even with this addition, the overall contribution of initial heterogeneity to lifetime hours inequality remains modest. Most of the dispersion in lifetime hours is generated after entry through

¹¹In the simulation, the conditional expectation $\mathbb{E}[Y_i | X_{i0}]$ is computed by grouping workers over the discretized state space. Continuous assets, human capital, and match quality are assigned to adjacent grid points using linear interpolation weights. Workers without an initial job are assigned to a separate no-match state. Preference types are already discrete.

Table 5: Sources of Lifetime Inequality

Statistic	From Entry			From Year 5
	Initial a_0, s_0	+ match m_0	+ prefs ξ_0	a_0, s_0, m_0, ξ_0
Lifetime earnings	0.032	0.145	0.264	0.456
Lifetime utility	0.317	0.372	0.492	0.613
Lifetime wealth	0.129	0.230	0.338	0.476
Lifetime hours	0.052	0.100	0.177	0.226

employment and preference shocks, job mobility, and the endogenous dynamic of labor-market states.

The last column of Table 5 uses the state vector five years after labor-market entry as the base period. The importance of initial conditions rises sharply. With the full state vector, the explained share of lifetime earnings inequality nearly doubles, from 26.4 to 45.6 percent. For lifetime wealth, the share rises from 33.8 to 47.6 percent; for lifetime utility, from 49.2 to 61.3 percent; and for lifetime hours, from 17.7 to 22.6 percent. Measuring “initial” conditions five years after entry, therefore, reassigns roughly half of the residual variance to initial heterogeneity for earnings and wealth.

The year-5 base period corresponds to age 23 for the average high-school graduate in our sample, the same chronological cutoff used by Huggett et al. (2011). At this comparable base period, our framework attributes 45.6 percent of lifetime earnings inequality to initial conditions, which is lower than the roughly 60 percent they report. Their framework features a Ben-Porath learning technology with endogenous time allocation between work and learning, but no leisure-consumption margin or frictional search – features that, in our framework, generate post-entry dynamics that reduce the apparent role of initial conditions. The 15-point gap between their estimate and ours, therefore, reflects modeling differences. The 19-point gap between our year-5 number and our entry number reflects measurement timing within the same framework. The latter gap means that a substantial share of lifetime inequality is produced after the school-to-work transition through channels that age-based decompositions with restricted specifications miss.

State variables observed several years after entry are not purely initial endowments. They already embed the effects of early-career shocks, labor-supply choices, human capital accumulation, and match sorting. Treating those states as initial conditions assigns part of the school-to-work transition to predetermined heterogeneity. Our framework allows us to separate the two, and the separation shows

that a substantial share of what age-based decompositions with restricted human-capital specifications attribute to initial heterogeneity is, in fact, produced during the early career. The next two subsections document the channels through which this happens.

5.2 Accounting for Wage and Earnings Growth and Dispersion

We next use the model to account for the sources of growth and dispersion in wages and earnings. The exercise is related to the decomposition in [Menzio et al. \(2016\)](#), who separate wage growth into the contributions of experience and match quality. We extend this logic in two directions. First, we apply the same accounting not only to mean wages but also to wage dispersion. Second, we study earnings, which adds hours worked as an additional source of both growth and inequality. The decomposition is a wage-equation accounting and therefore captures only the direct effect of each state on wages. As Subsection 5.3 will show, human capital also raises match quality indirectly, through the transition-rate and offer-quality channels – channels that this decomposition does not isolate. The human-capital and match-quality contributions reported below are therefore not fully independent: part of the match-quality contribution is itself produced by human capital operating through transitions and offers.

The decomposition is implemented using annual averages of monthly log components among employed workers. We average monthly logs, rather than taking logs of annual averages, because the additive wage and earnings decompositions hold at the monthly level in the simulated data.^{12 13}

¹²Given the function form that we assumed for the wage equation, mean log wages decompose into the contributions of human capital, match quality, and the age component:

$$\Delta \mathbb{E}[\log w_t] = \Delta \mathbb{E}[\rho_{ws} \log s_t] + \Delta \mathbb{E}[\rho_{wm} \log m_t] + \Delta \alpha_t.$$

Wage dispersion decomposes into variance and covariance components:

$$\begin{aligned} \text{Var}(\log w_t) &= \text{Var}(\rho_{ws} \log s_t) + \text{Var}(\rho_{wm} \log m_t) \\ &\quad + 2\text{Cov}(\rho_{ws} \log s_t, \rho_{wm} \log m_t). \end{aligned}$$

The first term captures dispersion in human capital, the second captures dispersion in match quality, and the covariance term captures sorting between the two. A positive covariance indicates that workers with higher human capital hold better matches, amplifying wage inequality.

¹³Earnings equal wages times hours, so log earnings decompose additively among employed workers:

$$\log e_{it} = \log w_{it} + \log h_{it},$$

and earnings dispersion satisfies

$$\text{Var}(\log e_t) = \text{Var}(\log w_t) + \text{Var}(\log h_t) + 2\text{Cov}(\log w_t, \log h_t).$$

Mean Growth Figure 6 reports the decomposition from labor-market entry. Mean wage growth reflects both learning and movement toward better matches. Over the first 20 years, mean log wages rise by 0.552 log points, with human capital contributing 0.326 log points and match quality contributing 0.322 log points. Wage growth comes from both forces, and they are quantitatively similar over the early career. Mean earnings grow more strongly, by 0.777 log points, because hours also rise: human capital and match quality each contribute about 0.32 log points, and hours contribute 0.225 log points. The contribution of hours to earnings growth is the wage-side counterpart of the dynamic labor-supply incentive documented in Subsection 5.3: early-career workers raise labor supply as they accumulate human capital and move into better matches, with the human capital impact on transition rates linking the three.

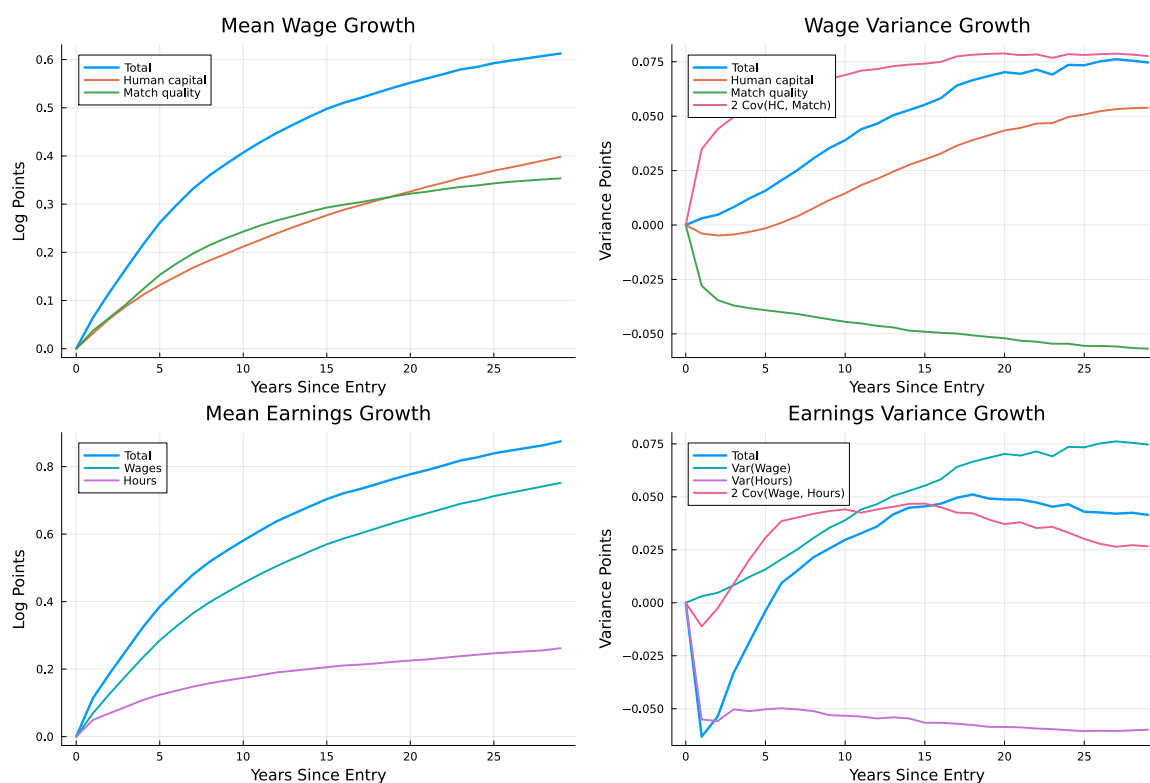


Figure 6: Wage and earnings growth and dispersion decompositions from labor-market entry.

Dispersion Growth Wage dispersion follows a different logic from mean growth. Variance rises from 0.108 at entry to 0.178 after 20 years, but the increase is not driven by widening match-quality dispersion, which actually falls over the career. The increase is driven by the rising covariance between human capital and match

This separates earnings dispersion into wage dispersion, hours dispersion, and the covariance between wages and hours.

quality, which moves from -0.46 at entry to positive after five years and accounts for most of the growth in wage variance. In words: workers with higher human capital increasingly hold better matches. Wage inequality grows because skills and jobs become aligned, not because either distribution widens in isolation.¹⁴

The sorting result reflects the wage-side channels documented in Subsection 5.3. Higher-human-capital workers face better offer distributions and lower separation risk, so skill accumulation translates endogenously into better matches. The positive covariance between s and m in the wage equation is the consequence of these channels operating on the joint state distribution. Without the transition-rate and offer-quality human-capital channels, the covariance would not rise: a wage-only specification of human capital cannot generate the sorting pattern that drives wage-inequality growth in the data.

Earnings dispersion is shaped by wage and hours dispersion. Wage dispersion rises from 37 percent of earnings variance at entry to 53 percent after 20 years, while hours dispersion falls from 44 percent to 21 percent. The wage-hours covariance is positive and rising, so high-wage workers also work longer hours, but the decline in hours dispersion dampens the increase in earnings inequality. The hours dispersion falls because the search-driven component declines, but it stabilizes at a level determined by preference heterogeneity. Without preferences, it would fall further. Earnings variance, therefore, rises less sharply than wage variance.

Timing of Measurement We now show how the timing of measurement shapes the interpretation of these decompositions. Table 6 compares the accounting when the base period is labor-market entry with the accounting when the base period is five years after entry. To keep the comparison sharp, the table reports changes over the next ten years from each base period.

The timing comparison reverses the interpretation of wage inequality growth. Measured from entry, the dominant force is sorting: the covariance between human capital and match quality contributes 176.8 percent of wage variance growth over the first ten years, while match-quality dispersion declines by 114.1 percent and human-capital dispersion contributes only 37.3 percent. Measured from year 5, the human-capital component contributes 80.2 percent, and the covariance contribu-

¹⁴The negative initial correlation between $\log s_0$ and $\log m_0 - \rho_{sm0} = -0.4592$ in the estimated model – is disciplined by observed dispersion in entry-period wages. The rise from negative to positive is the wage-side reflection of the transition-rate and offer-quality channels: higher-human-capital workers face better offer distributions and lower separation risk, so the joint distribution of s and m moves toward positive covariance even though the initial calibration starts it at negative covariance.

Table 6: Timing and the Accounting of Mean and Dispersion Growth

Component	From Entry		From Year 5	
	Contribution	Percent	Contribution	Percent
<i>Panel A: Mean wage growth</i>				
Human capital	0.212	52.1%	0.145	61.2%
Match quality	0.243	59.7%	0.140	59.0%
Total	0.407	100.0%	0.237	100.0%
<i>Panel B: Wage variance growth</i>				
Human capital	0.015	37.3%	0.032	80.2%
Match quality	-0.044	-114.1%	-0.010	-25.0%
Covariance	0.069	176.8%	0.018	44.8%
Total	0.039	100.0%	0.040	100.0%
<i>Panel C: Earnings variance growth</i>				
Wages	0.039	131.1%	0.040	80.1%
Hours	-0.053	-179.6%	-0.006	-12.8%
Covariance	0.044	148.4%	0.016	32.7%
Total	0.030	100.0%	0.049	100.0%

Note: The table compares ten-year changes using two different base periods. The entry-based columns measure changes from labor-market entry to year 10. The year-5 columns measure changes from year 5 to year 15 after entry. Panel A reports changes in mean log wages; Panels B and C report changes in variances. Percent contributions in Panel A are relative to total mean wage growth including the deterministic age component, which is not reported separately.

tion falls to 44.8 percent. Starting later, therefore, makes wage inequality growth look like a story of widening human-capital dispersion, while starting from entry reveals it as a sorting story driven by the channels through which human capital affects labor-market position.

The same pattern appears for earnings, moderated by hours. Wage dispersion accounts for 131.1 percent of earnings variance growth from entry but only 80.1 percent from year 5, while the offsetting role of hours dispersion shrinks from -179.6 to -12.8 percent. The dynamics that the entry-based decomposition seeks to explain are already partly embedded in the year-5 state distribution.

The implication for the broader inequality literature is the same as in Subsection 5.1. Decompositions that begin five years after entry treat the sorting between human capital and match quality, the convergence of employment states, and the early-career wage growth as predetermined, and miss the channels through which they were produced. Measuring workers from labor-market entry reveals that early-career sorting is itself a central source of wage inequality growth. We show in the next subsection that the sorting is generated by the transition-rate and

offer-quality channels through which human capital shapes labor-market position. This is one of the central reasons to track workers from entry rather than from a fixed age cutoff: the early career is when these channels do most of their work, and a decomposition that starts later conditions on outcomes the framework is trying to explain.

5.3 Mechanisms and Counterfactual Economies

The previous sections used the estimated model to decompose life-cycle profiles and lifetime inequality. We now use counterfactual economies to isolate the mechanisms behind those results and identify the forces that drive early-career labor supply. We compare the baseline economy to three counterfactuals, each corresponding to a canonical specification in the prior literature.

The first counterfactual is a standard life-cycle search economy with initial productivity and preference heterogeneity but no human-capital accumulation. This is the limiting case of frictional search models that abstract from endogenous skill. The second is a standard human-capital economy in which all workers start employed, the initial match is permanent, and human capital evolves through labor supply and affects wages alone. This is the limiting case of life-cycle labor-supply models with human-capital accumulation but no search frictions, in the tradition of Heckman (1976), Imai and Keane (2004), and Keane and Rogerson (2015).¹⁵ The third counterfactual shuts down preference heterogeneity to isolate its role in life-cycle hours dispersion, an issue emphasized by Bick et al. (2024).¹⁶

We further decompose the human-capital mechanism into three channels: wages, transition rates, and the mean of the offer distribution. As in standard life-cycle labor-supply models with human-capital accumulation, labor supply has an intertemporal return: work today raises future human capital, so the return

¹⁵Appendix C considers an additional portable-human-capital economy, in which accumulated human capital is no longer partly match-specific, and shows that allowing skills to travel across jobs strengthens rather than weakens the interaction between human capital and search.

¹⁶In the standard-search economy, we set the human-capital accumulation, depreciation, and loss parameters to zero, and remove human capital from job-finding rates, separation rates, and offer-quality distributions. Wages still depend on the initial level of human capital. The offer-quality intercepts and residual variances are adjusted to preserve the unconditional initial mean and variance of offer quality implied by the baseline model. The level of finding and separation rates is chosen so that the economy converges to the target employment rate. In the standard human-capital economy, all workers begin employed, on-the-job offers and separations are shut down, and the initial match is permanent. The initial match-quality distribution is renormalized to preserve initial average wages, so that the difference between this economy and the baseline reflects the absence of search rather than a change in entry-period wage levels. In the no-preference-heterogeneity economy, the initial and stationary dispersion in preferences is set to zero.

to working an additional hour is not only the current piece rate but also the expected future gains generated by skill accumulation (Heckman, 1976; Imai and Keane, 2004; Keane and Rogerson, 2015). In our framework, these future gains arise through three channels. The standard channel emphasized by the prior literature is that human capital raises future wages. The two additional channels are that human capital raises job-finding rates and lowers separation rates, and that it shifts the mean of the offer distribution upward, so that workers with more human capital draw from better job-quality distributions. We study each channel in isolation.¹⁷

Mechanism Comparison We start with Figure 7, which illustrates how the model’s identification works. The top-left panel reports mean wages. All economies generate rising wages, with the baseline producing the fastest and highest growth because it combines all mechanisms. The bottom-left panel reports wage dispersion and shows a sharper contrast: economies with human capital generate increasing dispersion, while the standard-search and no-preference-heterogeneity economies generate decreasing or flat dispersion. The standard-search economy converges to a stationary distribution of match quality because the high finding rate and low separation rate imply a flat dispersion for mid-career workers. Human capital is therefore essential for generating the observed increase in wage dispersion.

The top-right panel reports mean total hours – the product of the workweek and the employment rate. As with wages, all economies produce rising hours, but the path differs. The standard-search economy generates a sharp early-career increase from both margins: workers transition out of unemployment into employment, and rising match quality raises wages, which raises hours conditional on employment through the substitution effect. The increase flattens once the match-quality distribution stabilizes, since hours track wages through the substitution channel. The human-capital economy generates a smoother increase, since all workers are already employed and hours rise only through the workweek. The

¹⁷Each channel counterfactual starts from the baseline-calibrated parameters and shuts down a single channel. In the no-wage channel counterfactual, the wage elasticity with respect to human capital, ρ_{ws} , is set to zero, and the wage constant is renormalized to preserve the initial average wage. In the no-transition-rate channel counterfactual, the human-capital elasticities of the job-offer and separation rates, $\rho_{\lambda s}$ and $\rho_{\pi s}$, are set to zero, and their intercepts are adjusted to preserve average initial finding, job-to-job, and separation probabilities. In the no-offer-quality channel counterfactual, human capital is removed from the mean of the match-quality offer distributions, and the intercepts are adjusted to preserve average initial offer quality. The full no-human-capital-dynamics counterfactual shuts down accumulation, depreciation, and human-capital losses, while leaving the initial cross-sectional distribution of human capital and its static effects on wages, transition rates, and offer quality intact.

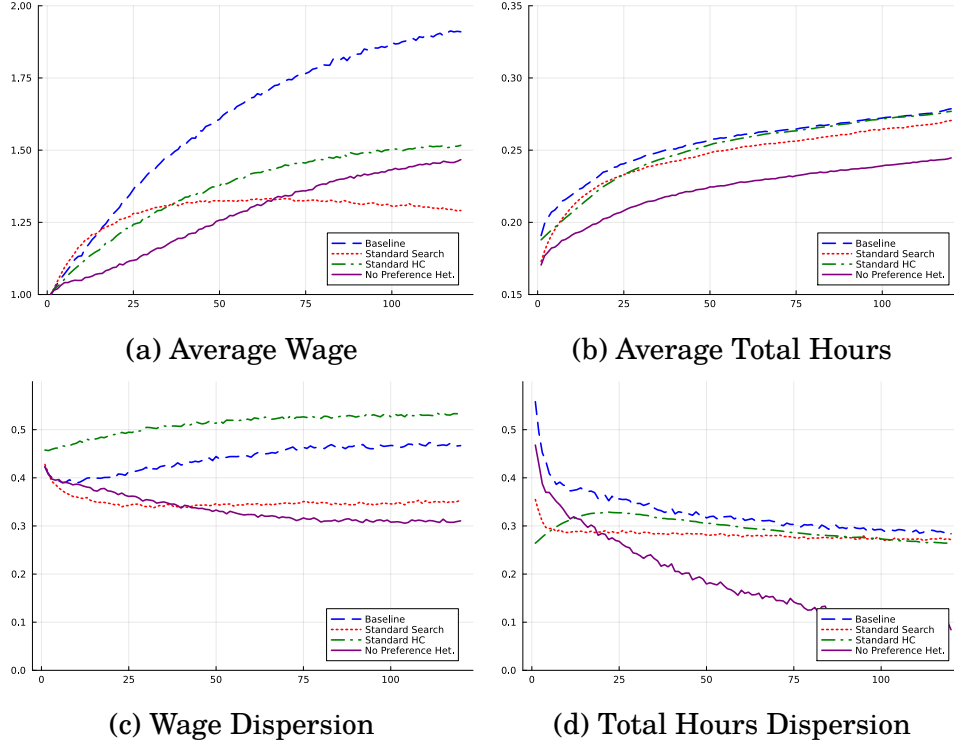


Figure 7: Counterfactual Economies

baseline tracks a path between the two, reflecting the contribution of both mechanisms. The no-preference-heterogeneity economy generates a similar path to the baseline but at a lower level, because removing the dispersion in preference types shifts the average disutility of work.

The bottom-right panel reports total hours dispersion. Search frictions are essential for generating the high early-career hours dispersion observed in the data, echoing the role of the wedges identified by [Kaplan \(2012\)](#). In the standard-search economy, hours dispersion mirrors wage dispersion through the same substitution channel: as the match-quality distribution converges, so does the hours distribution. The human-capital economy, by contrast, generates rising hours dispersion early in the career – the opposite of the data – because rising human capital raises wages, which induces a substitution toward more hours, and dispersion in human-capital paths translates into dispersion in hours. Preference heterogeneity is essential for generating persistent hours dispersion through the life cycle, echoing [Bick et al. \(2024\)](#): without it, hours dispersion falls substantially as workers’ employment and human-capital states converge.

We replicate the hours decomposition from Subsection 4.5 to understand how each mechanism contributes to the intensive and extensive margins of labor supply. Results are in Table 7 for 10 years after labor-market entry. In the baseline,

hours growth is almost evenly split between the two margins: 56.5 percent comes from higher hours conditional on employment, and 43.1 percent comes from higher employment. The two canonical economies differ on this split.

In the standard-search economy, the intensive margin accounts for 94 percent of hours growth – a standard substitution effect, as rising match quality raises the return to working an additional hour. The extensive-margin contribution is 21 percent, but it is partly offset by a negative composition term of -15 percent, so their net contribution is small. The negative composition term arises because workers in the decomposition window have, on average, longer tenure and therefore higher hours, while newly hired workers have lower hours on average. The no-preference-heterogeneity economy is close to the baseline, with a slight shift toward the intensive margin, and the standard-human-capital economy has only the intensive margin active by construction.

Table 7: Labor Supply Decomposition at Year 10

Economy	Intensive	Extensive	Composition	Quadratic
Baseline	56.5%	43.1%	0.2%	0.3%
Standard Search	93.9%	20.7%	-15.4%	0.8%
Standard HC	100.0%	0.0%	0.0%	0.0%
No Pref. Het.	63.8%	38.4%	-3.9%	1.7%

Both canonical economies depart from the empirical split in distinct ways. The standard-search economy generates an extensive-margin contribution that is too small once composition is netted out. The standard-human-capital economy excludes the extensive margin entirely. The baseline matches the data by combining both: search frictions generate employment growth, and human-capital accumulation generates workweek growth.

Another way of examining labor-supply incentives is to compute, for each employed worker, the ratio of the marginal rate of substitution between leisure and consumption to the current wage,

$$\frac{\text{MRS}_{it}}{w_{it}} = \frac{\xi_{it} h_{it}^{1/\gamma} c_{it}}{w(s_{it}, m_{it})}. \quad (5)$$

A static labor-supply model predicts that this ratio equals one, since workers choose hours where the marginal cost of work (MRS) equals the marginal benefit (the wage). The ratio exceeds one when workers supply hours at which the static cost of work exceeds the static benefit. The wedge summarizes the strength of dynamic

incentives – investment in future productivity, future labor-market prospects, and current resources when borrowing constraints limit consumption smoothing – that lead workers to choose hours beyond the static optimum. The presence of such a wedge is emphasized in [Keane and Rogerson \(2015\)](#), who argue that human-capital accumulation reconciles small static Frisch elasticities with the larger dynamic responsiveness implied by macro labor-supply behavior.

[Imai and Keane \(2004\)](#) plot this ratio to document the force of human-capital accumulation in their estimated model, reporting an MRS/wage ratio of 2.0 at age 20 that converges toward one over the life cycle. We do the same in Figure 8. The baseline and the human-capital economy both generate declining dynamic incentives over the life cycle, consistent with the pattern in Imai and Keane. The standard-search economy generates a flat ratio close to one, indicating that an economy without human-capital accumulation produces essentially no dynamic wedge: workers supply hours at the static optimum. The declining profile in the economies with human capital reflects two forces. The marginal return to additional hours falls with age because (i) the human-capital production function has decreasing returns, and (ii) the remaining lifespan decreases with age. The no-preference-heterogeneity economy generates a wedge that closely tracks the baseline, indicating that preference heterogeneity has little effect on the average dynamic incentive.

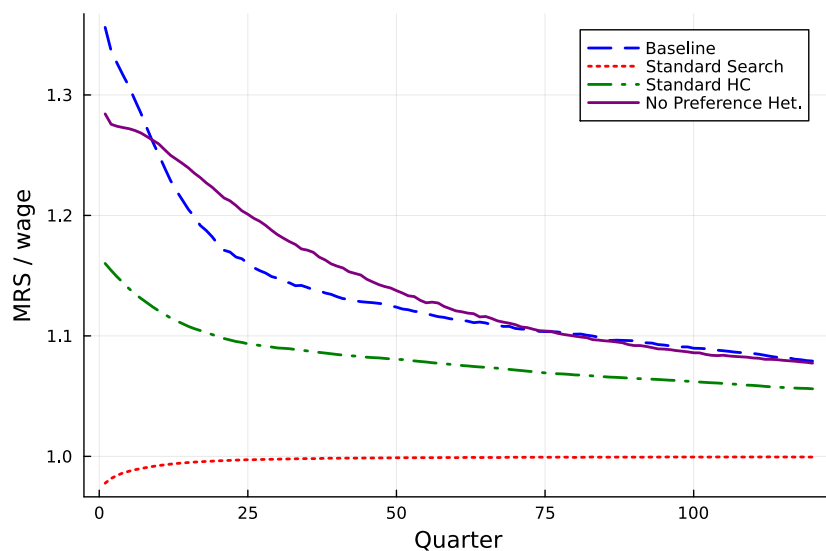


Figure 8: Marginal Rate of Substitution Relative to Wages

Table 8 quantifies the magnitudes. In the baseline, the average MRS/wage ratio is 1.253, 1.201, and 1.158 over the first five, ten, and twenty years. The remaining columns report the size of the wedge in each counterfactual relative to the baseline.

Table 8: Marginal Rate of Substitution: Baseline and Counterfactuals

Years	Baseline MRS/ w	Dynamic Wedge Relative to Baseline		
		Standard Search	Standard HC	No Pref. Het.
5	1.253	-3.56%	48.76%	100.47%
10	1.201	-2.82%	53.16%	109.12%
20	1.158	-2.11%	57.85%	108.48%

Note: The first column reports the average MRS/wage ratio in the baseline economy over the horizon indicated by each row. The remaining columns report, for each counterfactual economy, the dynamic wedge as a percentage of the baseline wedge, computed as $100 \times [((\text{MRS}/w)_{cf} - 1) / ((\text{MRS}/w)_{base} - 1)]$. A value of 100 indicates a wedge equal in size to the baseline wedge. Values above 100 indicate a larger wedge than in the baseline, while values close to zero indicate essentially no dynamic wedge.

Three results emerge. First, the standard-search economy generates essentially no dynamic wedge, and the MRS/wage ratio is essentially at one. Without human-capital accumulation, there is no return to working beyond the static optimum. Second, the standard-human-capital economy produces about half the baseline wedge – 49 percent in the first five years, rising to 58 percent at twenty years. The standard human-capital economy misses half the baseline wedge despite having full human-capital dynamics. The difference comes from the channels through which human capital affects labor-market outcomes beyond the wage. Third, the no-preference-heterogeneity economy produces a wedge essentially equal to the baseline, slightly larger at longer horizons.

Human Capital Channel Decomposition The preceding counterfactuals show that human capital is central for generating dynamic labor-supply incentives. We now decompose the human-capital mechanism into three channels: wages, transition rates, and the mean of the offer distribution. We do it by removing each channel separately, alongside a counterfactual that removes human-capital dynamics entirely while preserving initial human-capital heterogeneity. Figure 9 reports the MRS/wage ratio in each economy.

No single channel accounts for the baseline dynamic wedge. Removing human capital from wages substantially lowers the MRS/wage ratio, confirming that the standard wage channel is an important source of dynamic incentives. Removing human capital from transition rates also lowers the wedge, especially early in the life cycle: higher human capital raises future job-finding rates and lowers separation risk, providing an additional motive for current labor supply. Removing human capital from the mean of the offer distribution has a smaller but non-trivial

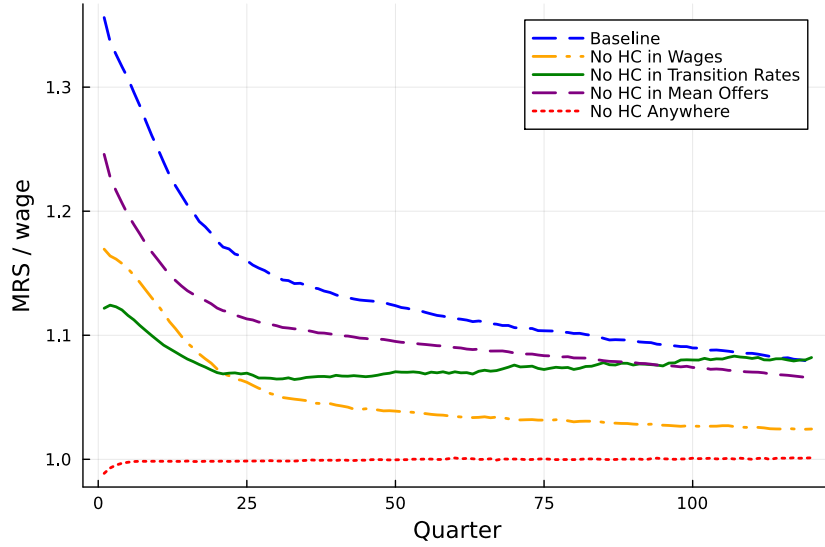


Figure 9: Marginal Rate of Substitution Relative to Wages

effect on the average wedge.

Table 9 quantifies the magnitudes. Under the same convention as Table 8, each column reports the wedge in the counterfactual as a percentage of the baseline wedge. Removing human capital from wages leaves 48 percent of the baseline wedge intact in the first five years and 39 percent at twenty years – equivalently, the wage channel accounts for roughly half the baseline wedge early in the career, with its share rising to nearly two-thirds at longer horizons. Removing human capital from transition rates leaves 38 percent of the wedge in the first five years, rising to 48 percent at twenty years. The transition-rate channel, therefore, accounts for slightly more than half the baseline wedge in the early career, with its share declining over time. Removing human capital from the mean of the offer distribution leaves 66 to 72 percent of the wedge, so the offer-quality channel accounts for the remaining 28 to 34 percent. Finally, the no-HC-dynamics economy produces an MRS/wage ratio essentially at one: without any human-capital accumulation, workers have no reason to supply hours beyond the static optimum.

Three patterns emerge from the channel decomposition. First, the wage channel and the transition-rate channel are each large and comparable in magnitude – both account for roughly half the baseline wedge in the early career, with their relative importance reversing over the life cycle. The two channels cross over in importance around the middle of the career: in the first decade the transition-rate channel dominates, and in the second decade the wage channel does. Second, the offer-quality channel is smaller but still meaningful: it accounts for about one-third of the dynamic wedge. Third, the channels are not additive substitutes. Each is

Table 9: Human Capital Channel Decomposition

Years	Baseline	Dynamic Wedge Relative to Baseline			
		No HC in Wages	No HC in Trans. Rates	No HC in Mean Offers	No HC Anywhere
5	1.253	47.93%	38.18%	65.90%	-1.05%
10	1.201	43.47%	40.62%	68.41%	-0.96%
20	1.158	38.91%	48.29%	72.13%	-0.64%

Note: The first column reports the average MRS/wage ratio in the baseline economy over the horizon indicated by each row. The remaining columns report, for each counterfactual economy, the dynamic wedge as a percentage of the baseline wedge, computed as $100 \times [((\text{MRS}/w)_{cf} - 1) / ((\text{MRS}/w)_{base} - 1)]$.

necessary for the full baseline wedge, and the framework's contribution comes from having all three operate simultaneously.

The hours decomposition in Table 10 shows how the channels map into intensive and extensive margins. Removing human capital from wages leaves the decomposition close to the baseline (59.6 / 41.3 vs. 56.5 / 43.1): both margins remain active because human capital still affects transitions and offers. Removing human capital from transition rates collapses the extensive margin from 43.1 to 9.2 percent and pushes the intensive margin to 91.1 percent. In this economy, human capital still raises wages, but it no longer improves employment stability or job-arrival prospects, so the dynamic incentive to work for the sake of better future jobs disappears, and hours growth becomes almost entirely intensive-margin. Removing human capital from mean offers leaves the intensive-extensive split close to the baseline (61.5 / 41.7), indicating that offer-quality effects shape the level of the dynamic wedge but not the margin on which it operates.

Table 10: Labor Supply Decomposition at Year 10

Economy	Intensive	Extensive	Composition	Quadratic
Baseline	56.5%	43.1%	0.2%	0.3%
No HC in Wages	59.6%	41.3%	-1.1%	0.2%
No HC in Trans. Rates	91.1%	9.2%	-6.2%	5.9%
No HC in Mean Offers	61.5%	41.7%	-3.3%	0.1%
No HC Anywhere	84.0%	6.0%	6.8%	3.2%

6 Conclusion

This paper studies the early career as a period of great importance for lifetime labor-market outcomes. We estimate a life-cycle model with an endogenous workweek, human-capital accumulation, and frictional labor markets, using profiles indexed by quarters since labor-market entry. The framework allows hours worked to shape human capital, and human capital to shape wages, job-finding rates, separation rates, and the quality of outside offers. Estimating from labor-market entry rather than a fixed age cutoff allows the school-to-work transition to be measured rather than discarded.

The estimated model yields three findings. First, the timing of measurement is decisive for how lifetime inequality is decomposed: initial conditions explain 26 percent of lifetime earnings dispersion when measured at labor-market entry, but 46 percent when measured five years after entry. This 20-percentage-point gap means that a substantial share of lifetime earnings inequality is produced during the school-to-work transition rather than inherited at its start. Second, wage inequality grows in the early career because human capital and match quality become increasingly aligned through the channels by which human capital shapes labor-market position; the rising covariance between the two accounts for most of the increase in wage variance. Third, human-capital accumulation is the dominant force behind early-career labor supply, with roughly two-thirds of its importance operating through transition rates and outside-offer channels rather than through wages alone. Together, the findings support the conclusion that modeling the labor-market churn and human-capital channels of the early career is essential for understanding lifetime labor-market outcomes.

Two extensions seem especially relevant. First, our model abstracts from heterogeneity in education and family structure. Incorporating these dimensions would allow the framework to speak more directly to differences across demographic groups, as in [Peditke \(2026\)](#), who studies the intergenerational implications of family structure. Second, we model hours as a continuous choice, while in practice, workers often face fixed-hour contracts, part-time arrangements, and other institutional constraints on adjustment along the intensive margin. Assessing the implications of part-time arrangements would test the robustness of our dynamic-hours incentive conclusion, particularly given that [Gimpelson \(2024\)](#) shows these constraints account for a substantial share of the variation in lifetime hours in Germany.

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A Additional Empirical Results

A.1 Sample Description

Table A1: Descriptive Statistics

	Full Cohort	HS Graduates	Final Sample
Age (quarters)	133.92 (50.29)	137.94 (51.24)	147.35 (52.09)
Sex	1.51 (0.50)	1.46 (0.50)	1.47 (0.50)
Highest grade ever	14.01 (3.15)	11.96 (0.19)	11.99 (0.09)
Total hours worked	377.29 (1351.91)	361.03 (1170.26)	373.73 (1228.60)
Real hourly wage	12.56 (8.44)	10.74 (6.36)	11.02 (6.46)
Fraction unemployed	0.05 (0.19)	0.06 (0.21)	0.06 (0.21)
Fraction employed	0.70 (0.44)	0.66 (0.45)	0.67 (0.45)

A.2 Comparison Between the Two Cohorts

Figure A1 compares the mean profiles for the NLSY79 and NLSY97 cohorts. The main life-cycle patterns are similar across cohorts. Average wages are higher for the NLSY97 cohort than for the NLSY79 cohort, while the average workweek is lower for the NLSY97 cohort. By contrast, employment rates and the cumulative number of jobs held are very similar across cohorts.

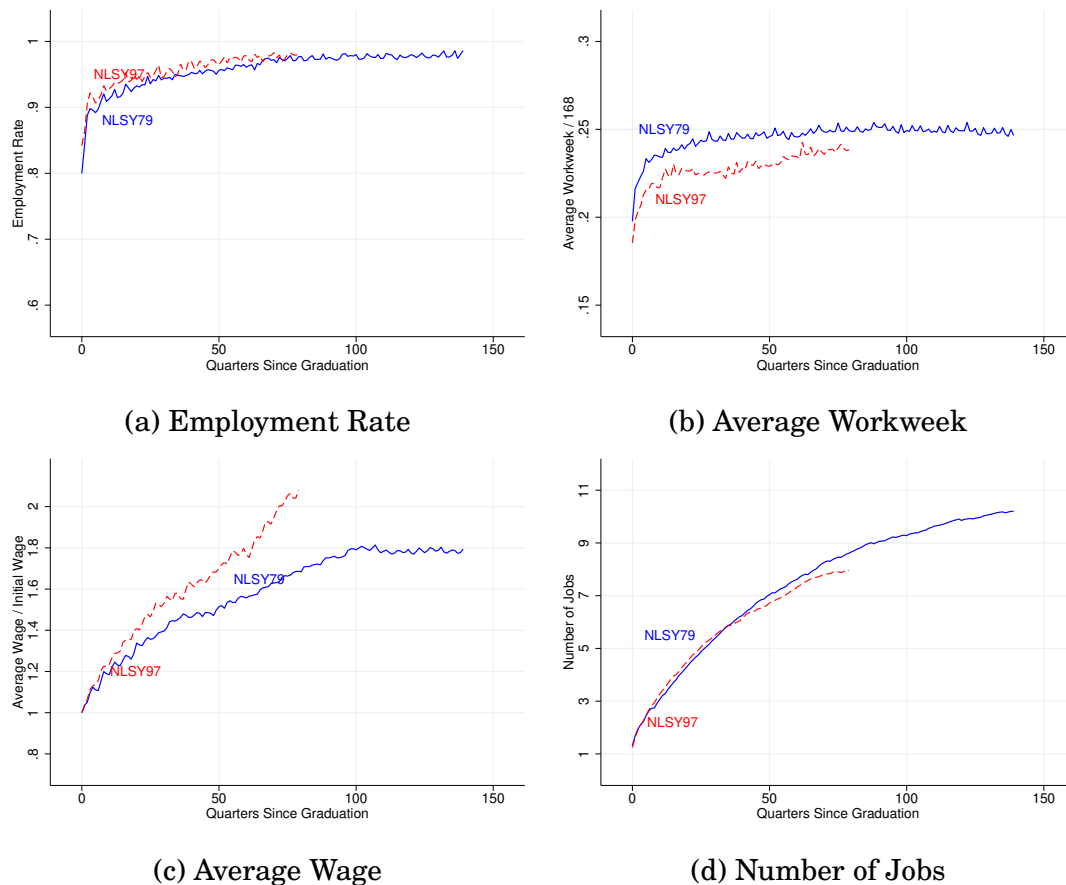


Figure A1: Mean Profiles by Cohort

Note: The figure compares mean profiles for the NLSY79 and NLSY97 cohorts. Employment is measured as the fraction of the quarter worked. The average workweek is normalized by 168 hours. Wages are normalized by the first observed wage in the profile. The number of jobs is measured as the cumulative number of distinct jobs held by the end of each quarter.

Figure A2 compares dispersion profiles across cohorts. The dispersion patterns are also highly similar. The standard deviations of the employment rate, average workweek, average wage, and cumulative number of jobs all track each other closely across the NLSY79 and NLSY97 cohorts. Thus, although the two cohorts differ in the levels of wages and workweeks, they display similar cross-sectional dispersion in the restricted sample.



Figure A2: Dispersion Profiles by Cohort

Note: The figure compares dispersion profiles for the NLSY79 and NLSY97 cohorts. The workweek and wage dispersion profiles are standard deviations of logs. Employment-rate dispersion is the standard deviation of the fraction of the quarter worked. Job dispersion is the standard deviation of cumulative jobs held.

Figure A3 shows that labor-market transition rates are also similar across cohorts. Job-to-job transitions, job-to-unemployment transitions, and unemployment-to-job transitions follow comparable life-cycle profiles in the NLSY79 and NLSY97. This reinforces the view that the main labor-market dynamics are stable across cohorts.

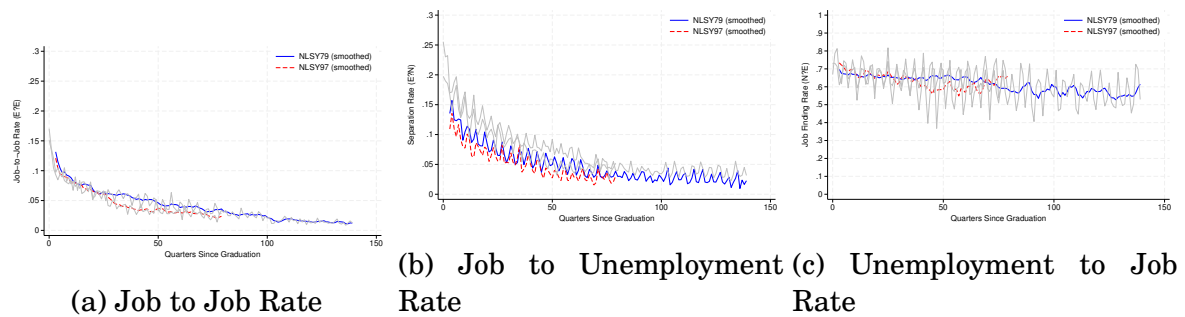


Figure A3: Transition Rate Profiles by Cohort

Note: The figure compares labor-market transition rates for the NLSY79 and NLSY97 cohorts. Transition rates are plotted by quarters since graduation.

A.3 Profiles Without Hours Restrictions

Figure A4 repeats the comparison without imposing the annual-hours restriction. The same broad differences across cohorts remain. Average wages are higher for the NLSY97 cohort, while the average workweek is lower. Employment rates and the cumulative number of jobs held remain similar across cohorts. Thus, removing the hours restriction does not materially change the comparison of mean profiles.

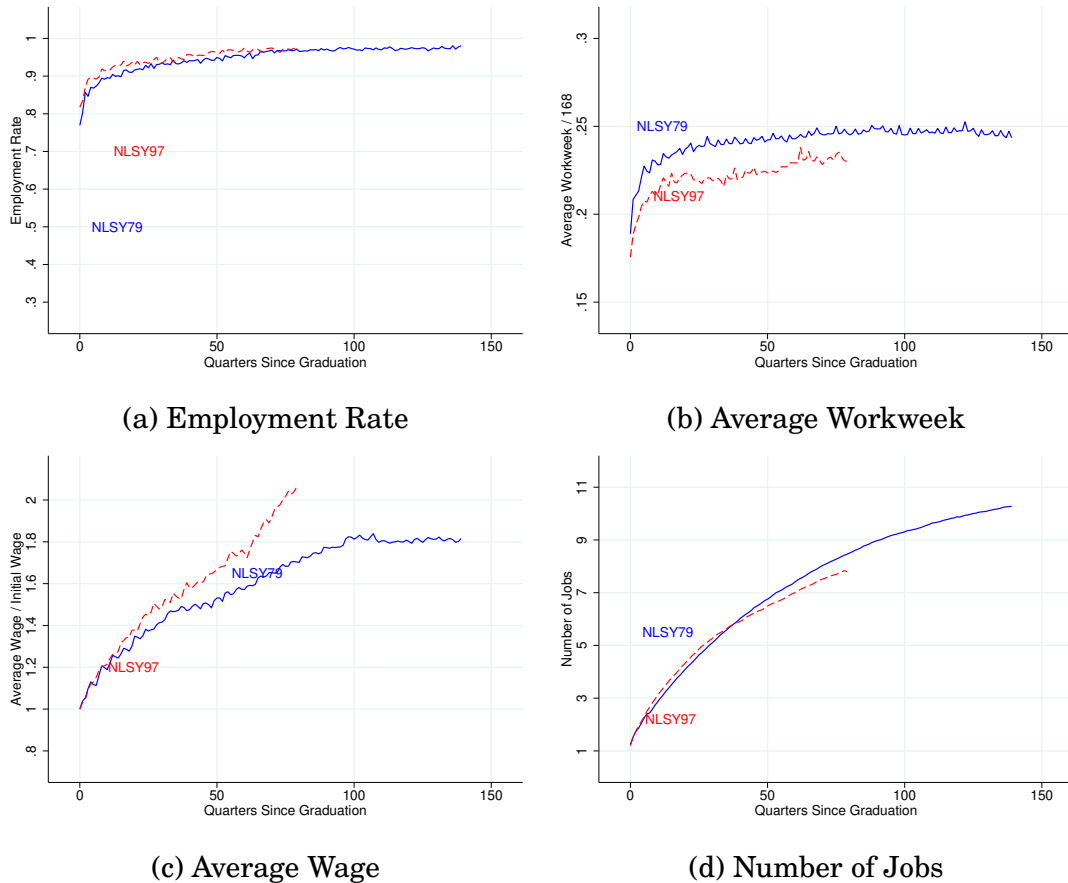


Figure A4: Mean Profiles Without Hours Restrictions

Note: The figure plots mean profiles without restricting the sample to individuals with annual hours between 520 and 5,200. The NLSY79 is represented by solid lines and the NLSY97 by dashed lines.

Figure A5 shows a more interesting pattern for dispersion. Employment-rate dispersion, wage dispersion, and dispersion in cumulative jobs remain similar across cohorts. By contrast, workweek dispersion is higher for the NLSY97 cohort than for the NLSY79 cohort. This differs from Figure A2, where workweek dispersion is similar across cohorts in the restricted sample. The comparison suggests that the Kaplan (2012)-style pattern in workweek dispersion is stable across cohorts among attached workers, but changes once less attached workers are included. Hence, the cohort difference in unrestricted workweek dispersion appears to be driven by workers with weaker labor-market attachment.

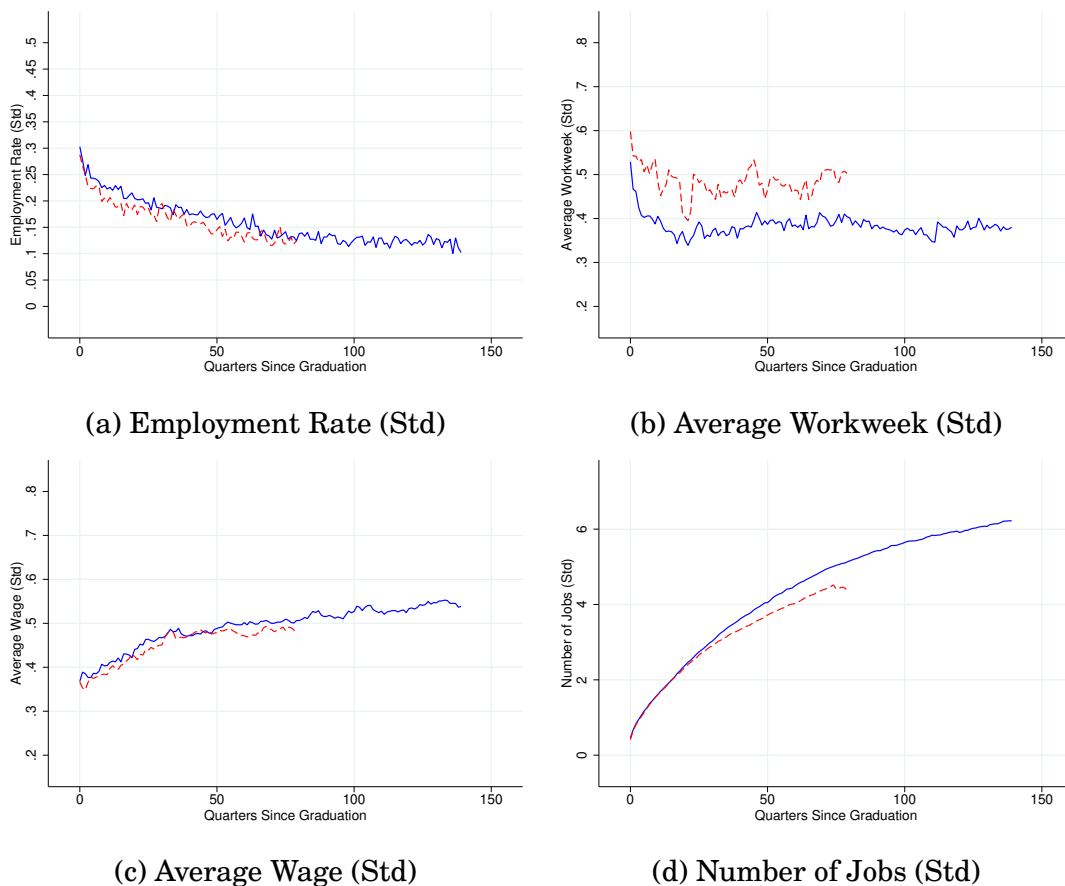


Figure A5: Dispersion Profiles Without Hours Restrictions

Note: The figure plots dispersion profiles without restricting the sample to individuals with annual hours between 520 and 5,200. The workweek and wage dispersion profiles are standard deviations of logs.

B Derivation of the Hours Decomposition

In this appendix, we derive the statistical decomposition that we use to quantify the relative importance of both the average number of weeks worked annually and the average weekly hours in accounting for the increase in total hours.

Goal: Derive a decomposition of total hours worked into “intensive” and “extensive” margins. The intensive margin is measured by the workweek, while the extensive margin is measured by The intensive margin is measured by the workweek, the within-continuers extensive margin is measured by weeks worked, and a separate composition term captures entry, exit, and changes in sample weights..

Idea: What would be the total hours worked if the intensive margin (or the extensive margin) is held constant? Perform a discrete-time approximation of the total derivative.

Notation:

- H_t = the average total hours in period t , where period t is measured in years since graduation. Its definition is: $H_t = \sum_{i \in E_t} \omega_{it} h_{it} n_{it}$.
- E_t = the set of all workers employed in period t .
- ω_{it} is the sample weight of worker i in period t . In the unweighted case, $\omega_{it} = 1/N_t$.
- h_{it} = the average workweek in that period for worker i .
- n_{it} = the number of weeks worked in that period by worker i .

Implementation issues:

- Because the sample is not a balanced panel, the sample weights change across periods. Changes in the sample composition, including workers’ entry, exit, and weight changes, need to be tracked.
- By performing a discrete approximation of the total derivative, second-order terms do not cancel out. They need to be tracked.
- Also, because of the approximation of the total derivative, changes need to be evaluated in a base period. The Laspeyres index is used, i.e., period $t - 1$ is the base period.

Derivation Sketch:

The derivation proceeds by following these steps:

- First, decompose $H_t - H_{t-1}$ into the intensive and extensive margins.
- Second, sum $H_t - H_{t-1}$ for $t \in \{1, \dots, T\}$ to find the contributions of the intensive and extensive margins over an extended period.

Formal Derivation:

A change in average total hours between periods t and $t - 1$, $H_t - H_{t-1}$, can be decomposed as follows:

$$\begin{aligned}
H_t - H_{t-1} &= \sum_{i \in E_t} \omega_{it} h_{it} n_{it} - \sum_{i \in E_{t-1}} \omega_{it-1} h_{it-1} n_{it-1} \\
&= \sum_{i \in E_t} \omega_{it} h_{it} n_{it} - \sum_{i \in E_{t-1}} \omega_{it-1} h_{it-1} n_{it-1} \pm \sum_{i \in E_{t \cap t-1}} \omega_{it} h_{it-1} n_{it-1} \\
&= \sum_{i \in E_{t \cap t-1}} \omega_{it} (h_{it} n_{it} - h_{it-1} n_{it-1}) \\
&\quad + \underbrace{\sum_{i \in E_{t \cap t-1}} (\omega_{it} - \omega_{it-1}) h_{it-1} n_{it-1} + \sum_{i \in E_t \setminus E_{t-1}} \omega_{it} h_{it} n_{it} - \sum_{i \in E_{t-1} \setminus E_t} \omega_{it-1} h_{it-1} n_{it-1}}_{\text{composition change}} .
\end{aligned}$$

The last terms capture the impact of weight changes, as well as entry and exit. We bunch them together in a *composition change* term. The set $E_{t \cap t-1}$ is composed of all workers that are present in the sample in both periods t and $t - 1$, or, more formally, $E_{t \cap t-1} = \{i \mid i \in E_t \cap E_{t-1}\}$.

Continuing with the derivation, $H_t - H_{t-1}$ can be further decomposed as:

$$\begin{aligned}
H_t - H_{t-1} &= \sum_{i \in E_{t \cap t-1}} \omega_{it} (h_{it} n_{it} - h_{it-1} n_{it-1}) + \text{comp. change} \\
&= \sum_{i \in E_{t \cap t-1}} \omega_{it} (h_{it} n_{it} - h_{it-1} n_{it-1}) \pm \sum_{i \in E_{t \cap t-1}} \omega_{it} h_{it} n_{it-1} + \text{comp. change} \\
&= \sum_{i \in E_{t \cap t-1}} \omega_{it} (h_{it} - h_{it-1}) n_{it-1} + \sum_{i \in E_{t \cap t-1}} \omega_{it} h_{it} (n_{it} - n_{it-1}) + \text{comp. change} \\
&= \sum_{i \in E_{t \cap t-1}} \omega_{it} (h_{it} - h_{it-1}) n_{it-1} + \sum_{i \in E_{t \cap t-1}} \omega_{it} h_{it} (n_{it} - n_{it-1}) \\
&\quad \pm \sum_{i \in E_{t \cap t-1}} \omega_{it} h_{it-1} (n_{it} - n_{it-1}) + \text{comp. change} \\
&= \underbrace{\sum_{i \in E_{t \cap t-1}} \omega_{it} (h_{it} - h_{it-1}) n_{it-1}}_{\text{intensive margin}} + \underbrace{\sum_{i \in E_{t \cap t-1}} \omega_{it} h_{it-1} (n_{it} - n_{it-1})}_{\text{extensive margin}} \\
&\quad + \underbrace{\sum_{i \in E_{t \cap t-1}} \omega_{it} (h_{it} - h_{it-1}) (n_{it} - n_{it-1})}_{\text{quadratic term}} + \text{comp. change} .
\end{aligned}$$

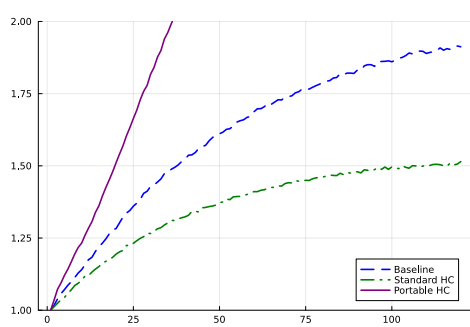
The first term refers to the intensive margin. It answers how much the increase in the workweek between t and $t - 1$ increased total hours worked, holding the number of weeks worked in a year constant at its $t - 1$ value. The second term refers to the extensive margin. Similarly, it answers how much the increase in weeks worked in a year between t and $t - 1$ increased total hours worked, holding the workweek constant at its $t - 1$ value. The quadratic term captures second-order changes. It is positive when weeks worked and the workweek tend to move together.

C The Role of Match-Specific Human Capital

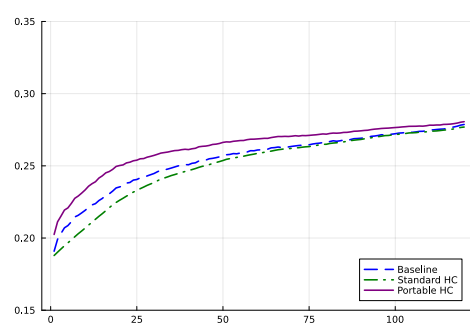
This appendix evaluates the role of match-specific human capital. The main text compares the baseline economy to canonical search and human-capital benchmarks. Here we ask a related question: if human capital is portable across jobs, does a model with human capital and search resemble the canonical standard human-capital model?

In the baseline model, human capital is partly match-specific. Workers lose a fraction of their accumulated skills when they separate or change jobs, as captured by the loss function $\phi(s) = Bs^\nu$. We construct a portable-human-capital counterfactual by shutting down these losses and removing match quality from the human-capital production function. Human capital remains productive in wages, job-finding rates, separation rates, and the mean of the offer distribution, but it is no longer tied to the current match. We compare this economy to the baseline and to the standard human-capital economy, in which all workers start employed, the initial match is permanent, and human capital affects wages alone.

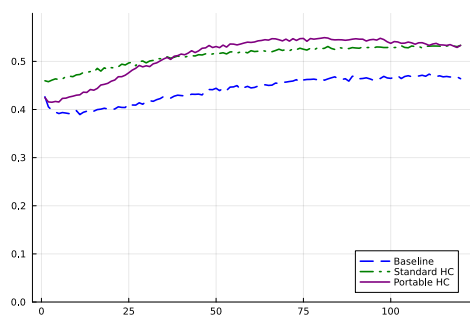
Figure C1 reports the life-cycle profiles. The portable-human-capital economy does not collapse to the canonical standard-human-capital benchmark. Instead, portability strengthens the interaction between human capital and search. Because workers can carry accumulated skills across jobs, current labor supply raises not only future wages but also future labor-market prospects without the offsetting risk of losing match-specific capital when moving. This raises the return to early-career work and changes both wage and hours dynamics.



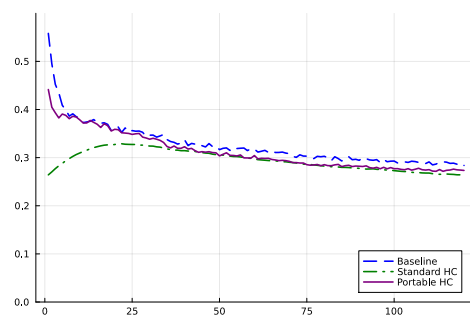
(a) Average Wage



(b) Total Hours



(c) Average Wage (Std)



(d) Total Hours (Std)

Figure C1: Human-Capital Portability

Figure C2 reports the marginal rate of substitution relative to wages. The portable-human-capital economy generates a substantially larger dynamic wedge than the baseline. This is the central result of the exercise. Portability makes human-capital investment more valuable because workers keep the returns to skill accumulation when they move up the job ladder. In this sense, search does not merely add an extensive margin to the standard human-capital model; it changes the return to human-capital investment itself.

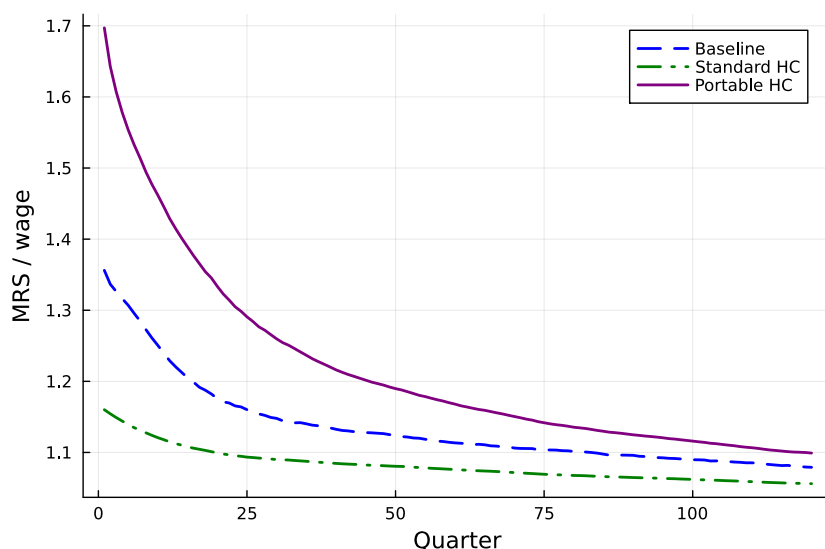


Figure C2: Marginal Rate of Substitution Relative to Wages

Table C1 quantifies the size of the dynamic wedge. The standard-human-capital economy produces about one-half of the baseline wedge, as in the main text. By contrast, the portable-human-capital economy produces a wedge that is 186 percent of the baseline over the first five years, 182 percent over the first ten years, and 170 percent over the first twenty years. The wedge is therefore not only larger than in the standard-human-capital benchmark, but also larger than in the baseline.

Table C1: Human Capital Portability

Years	Baseline MRS/ w	Dynamic Wedge Relative to Baseline	
		Standard HC	Portable HC
5	1.253	48.76%	186.42%
10	1.201	53.16%	182.45%
20	1.158	57.85%	169.80%

The hours decomposition in Table C2 shows that portability also changes the margin through which labor supply grows. In the standard-human-capital economy, all growth comes from the intensive margin by construction. In the portable-human-capital economy, the split is almost even: 49.4 percent of hours growth comes from higher hours conditional

on employment, and 50.1 percent comes from higher employment. Thus, portable human capital with search does not look like the canonical standard-human-capital model.

Table C2: Labor Supply Decomposition at Year 10: HC Portability

Economy	Intensive	Extensive	Composition	Quadratic
Baseline	56.50%	43.10%	0.15%	0.25%
Standard HC	100.00%	0.00%	0.00%	0.00%
Portable HC	49.41%	50.12%	-0.89%	1.37%

The conclusion is that match-specificity disciplines the dynamic incentive to work. When human capital is portable, workers retain a larger share of the return to skill accumulation when changing jobs. The result is an economy with a much larger dynamic wedge than the baseline and an extensive-margin contribution that remains central. Search with portable human capital, therefore, does not reduce to the canonical standard-human-capital model; it generates stronger incentives than either benchmark because human-capital accumulation and job search reinforce one another.